

Functional Specification

Version 2 · with UML models and UI mockups in place

DOCUMENT

Specification · UML · UI

DATE

2026-07-08

SCOPE

MVP, V1 & V2 requirements

STATUS

Requirement IDs are authoritative

Requirement IDs (FONC- / CONS- / PERF-) are stable and normative. UML diagrams and UI mockups illustrate the requirements; where they differ, the requirement text governs. UI mockups follow the project design charter (a light deep-time cartographic system).

Contents

1. Product vision
2. Functional specification
 1. Conventions & glossary
 2. Features — with domain model, use case, activity, state & sequence UML and UI mockups
 3. Constraints
 4. Performance
 5. MVP scope & out-of-scope
3. Data architecture & model (design) — sources (PBDB + Wikipedia), the three data tiers, the assertion/provenance pattern & typed class models
4. Appendix A — Requirements index
5. Appendix B — Requirements traceability
6. Appendix C — Acceptance criteria
7. Appendix D — Glossary

Figures are numbered sequentially. UML is rendered with Mermaid (typed class models with composition/aggregation, plus use case, activity, state & sequence diagrams); UI mockups are the project's high-fidelity SVG screens. Decisions on earlier ambiguities are folded into the requirements themselves.

Product vision

Derived document. It summarizes intent; it does not introduce requirements. Authoritative requirements live in [functional-specification.md](#).

One-line vision

A map-first tool for exploring **where** and **when** non-avian dinosaurs are known from fossil evidence across the Mesozoic era — scientifically cautious, source-driven, and honest about uncertainty.

What the product is

The atlas is built around an interactive **paleogeographic map** paired with a **geological time control**. As the user moves through geological time, the map redraws the ancient positions of the continents and shows the fossil occurrences that overlap the selected age. From any occurrence the user can open a **taxon profile** to see classification, time range, discovery locations, and sources.

The experience deliberately combines three dimensions:

- **Geography** — reconstructed paleogeographic positions plus modern discovery locations.
- **Time** — the Mesozoic interval (≈ 252 –66 Ma), split into Triassic, Jurassic, and Cretaceous.
- **Taxonomy** — exploration and filtering by taxonomic group and taxon.

Focus and content hierarchy

- **Non-avian dinosaurs are the primary content.** They are the default active category and the center of the experience.
- **Other Mesozoic reptiles are secondary content** (e.g. pterosaurs, marine reptiles). They are clearly labeled as secondary and can be hidden to preserve a dinosaur-centered view.
- The atlas does **not** claim to cover all Mesozoic life. It is not a complete atlas of plants, invertebrates, mammals, fish, or microorganisms.

Guiding principle: scientific caution

The product's identity depends on being trustworthy:

- Every visible fossil occurrence and time range is tied to an **identifiable source**.
- **Fossil points are evidence of discovery**, not claims about exact where an animal lived, and never an extrapolated distribution range.
- **Reconstructed** paleogeographic positions and **approximate** time ranges are labeled as such.
- **Interpretative data** (diet, mass, behavior) is distinguished from direct fossil facts, and **missing data is shown explicitly**, never silently filled with an unmarked assumption.
- The tone is informative, scientific, and accessible — not sensationalist and not overly childish.

Who it is for

Curious explorers who want a credible, visual way to understand the geography and timeline of dinosaurs — from enthusiasts to students and educators — without being misled by oversimplified or unsourced claims.

What success looks like (MVP)

A user can explore the Mesozoic, see a paleogeographic map for a chosen age, filter to dinosaurs, select a fossil occurrence, open its taxon profile, and return to the map — with sources, uncertainty, and missing data always visible. See [mvp-scope.md](#).

Functional specification

Authoritative source of product requirements. This is the single source of truth for what the atlas must do. Derived documents (indexes, traceability, vision, scope, glossary) summarize this file and must not contradict it. If a derived document conflicts with this specification, this specification wins.

Requirement IDs are **stable** and must not be renumbered. Priorities ([MVP], [V1], [V2]) and requirement verbs (**must** vs **should**) are normative and must be preserved.

0. Conventions

0.1 Priorities

Requirements are marked with a priority:

- **[MVP]**: required for a first usable version.
- **[V1]**: desirable for a complete first version, but not required for the MVP.
- **[V2]**: advanced feature or later enhancement.

0.2 Interpretation of requirement verbs

- **must**: mandatory requirement.
- **should**: optional or recommended requirement.

Clarification (priority vs. verb). Priority (§0.1) and verb are two independent axes: the **priority** says *which release* a requirement belongs to (MVP / V1 / V2), and the **verb** says *how binding* it is *within that release*. Consequently a "[V1] must" requirement is not required for the MVP but becomes mandatory once V1 is undertaken. For example, showing a taxon image is optional (FONC-1190, "[V1] should"), but *if* an image is shown, labeling its type, credit, and artistic nature is mandatory (FONC-1200–FONC-1230, "[V1] must"). This note documents interpretation only; it changes no requirement wording or ID.

0.3 Glossary

Mesozoic: geological interval between approximately 252 Ma and 66 Ma.

Ma: millions of years before present.

Non-avian dinosaurs: dinosaurs in the common sense, excluding modern birds and strictly avian lineages.

Taxon: biological classification unit, for example clade, family, genus or species.

Fossil occurrence: documented fossil evidence associated with a taxon, a discovery location and a geological time range.

Geological formation: named geological unit associated with rocks and, in some cases, fossils.

Modern position: present-day location where a fossil was discovered.

Paleogeographic position: approximate reconstructed position of that location during a past geological period.

Time range: interval between a minimum age boundary and a maximum age boundary.

Interpretative data: inferred or synthesized information, for example diet, estimated mass or assumed behavior.

Sourced data: information linked to an identifiable source.

Geological stage: a subdivision of a geological period in the International Chronostratigraphic Chart. The time control's selectable resolution is the geological stage (see §1.2).

Action: one primary user interaction that causes a navigation or state transition — a click/tap or its keyboard equivalent. Hover, scroll, zoom and pan are not counted. This is the unit used by the navigation limits in FONC-1070 / FONC-1080 and CONS-460 / CONS-470 (for example, occurrence → occurrence panel → taxon profile is two actions).

1. Features

1.1 Main exploration view

- **FONC-010** [MVP] — The system must provide a main exploration view combining a map, a time control and filters.
- **FONC-020** [MVP] — The system must display dinosaurs as the active category by default when the main exploration view first loads.
- **FONC-030** [MVP] — The system must allow access from the main exploration view to a fossil occurrence, a taxon page and a filter panel.
- **FONC-040** [MVP] — The system must permanently display the selected geological age on the main exploration view.
- **FONC-050** [MVP] — The system must permanently display the currently selected taxonomic group on the main exploration view.
- **FONC-060** [MVP] — The system must permanently display the number of visible occurrences matching the active filters.
- **FONC-070** [V1] — The system should display a short textual summary of the current selection, including the period, taxonomic group and number of results.
- **FONC-080** [MVP] — The system must allow the active filters to be reset from the main exploration view.

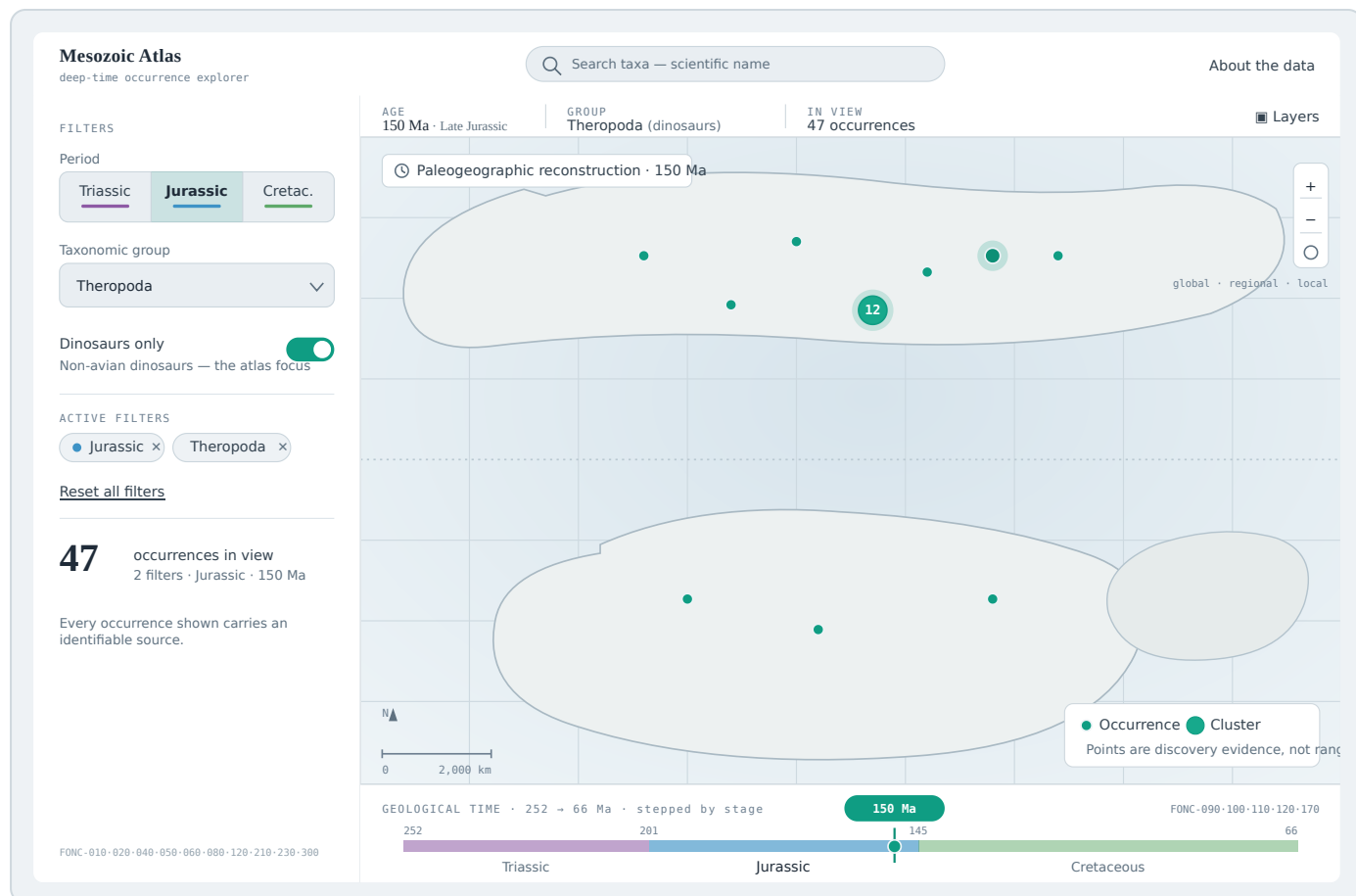


Figure 1 — UI mockup: Exploration view

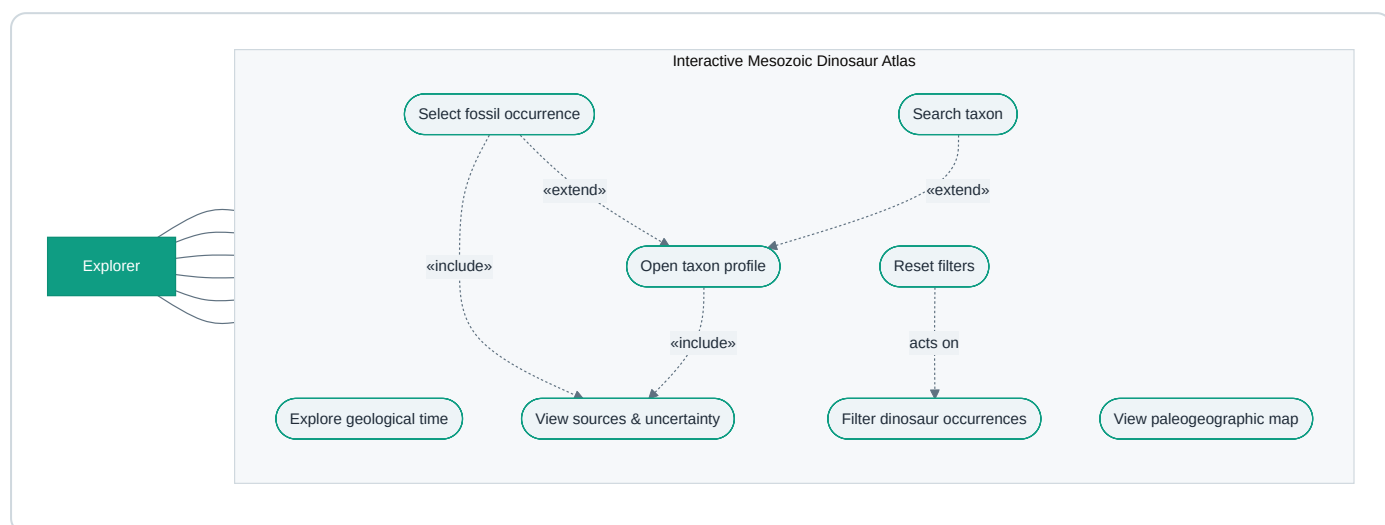


Figure 2 — Use case diagram — Explorer and the top-level use cases

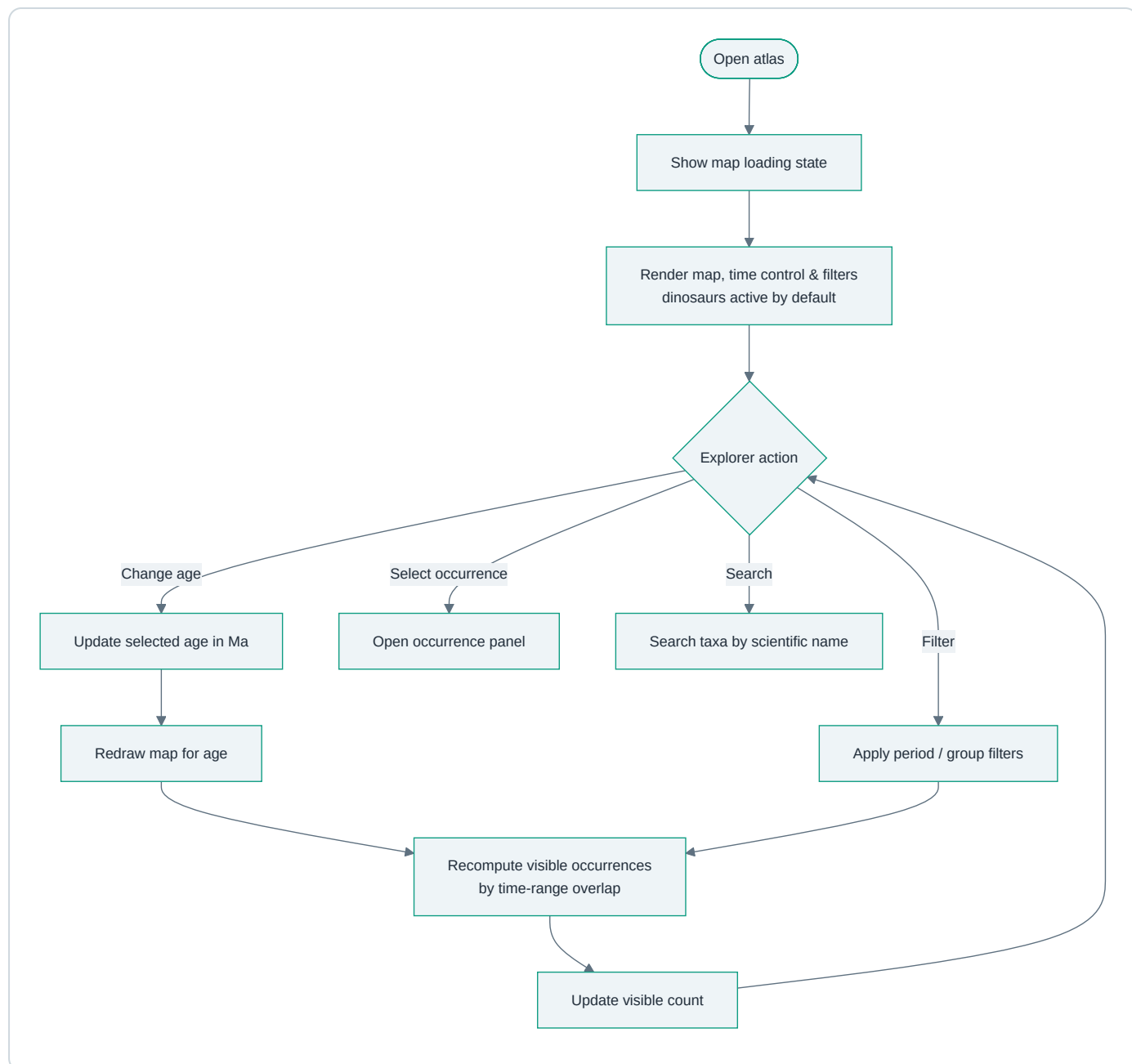


Figure 3 — Activity — main exploration & age change

1.2 Time exploration

Clarification (time control resolution). The selected age is chosen in discrete steps at **geological-stage** boundaries and is displayed in Ma; quick selection of a whole period (FONC-190) is a shortcut over the same control. This fixes how FONC-120 and FONC-170 behave and how the overlap between the selected age and a known time range (FONC-150 / FONC-160) is evaluated. A continuous, sub-stage age selector is out of scope.

- **FONC-090** [MVP] — The system must allow exploration of the interval between 252 Ma and 66 Ma.
- **FONC-100** [MVP] — The system must divide the main time interval into three periods: Triassic, Jurassic and Cretaceous.
- **FONC-110** [MVP] — The system must display the selected age in millions of years before present using the unit Ma.
- **FONC-120** [MVP] — The system must allow the selected age to be changed through a time control visible on the main exploration view.

- **FONC-130** [MVP] — The system must update the map when the selected age changes.
- **FONC-140** [MVP] — The system must update the visible fossil occurrences when the selected age changes.
- **FONC-150** [MVP] — The system must display an occurrence only when the selected age overlaps with the known time range of that occurrence.
- **FONC-160** [MVP] — The system must display a taxon only when the selected age overlaps with its known time range.
- **FONC-170** [MVP] — The system must visually distinguish a selected precise age from a known time range.
- **FONC-180** [V1] — The system should display the geological stage corresponding to the selected age when this information is available.
- **FONC-190** [V1] — The system should allow quick selection of a complete period, for example Triassic, Jurassic or Cretaceous.
- **FONC-200** [V2] — The system should allow two geological ages to be compared side by side.

1.3 Paleogeographic map

- **FONC-210** [MVP] — The system must display a world map corresponding to the selected age.
- **FONC-220** [MVP] — The system must show the approximate position of continental landmasses for the selected age.
- **FONC-230** [MVP] — The system must display fossil occurrences as points or grouped points on the map.
- **FONC-240** [MVP] — The system must visually distinguish individual occurrences from occurrence groups.
- **FONC-250** [MVP] — The system must allow the user to zoom on the map.
- **FONC-260** [MVP] — The system must allow the user to pan the map horizontally and vertically.
- **FONC-270** [MVP] — The system must allow a fossil occurrence displayed on the map to be selected.
- **FONC-280** [MVP] — The system must display an information panel when a fossil occurrence is selected.
- **FONC-290** [MVP] — The system must display at minimum, for a selected occurrence: the associated taxon, the time range, the modern location, the paleogeographic position and the source.
- **FONC-300** [MVP] — The system must clearly indicate that the ancient map displayed is a paleogeographic reconstruction.
- **FONC-310** [V1] — The system should allow a comparison layer with present-day continents to be displayed.
- **FONC-320** [V1] — The system should allow fossil occurrence display to be enabled or disabled.
- **FONC-330** [V2] — The system should allow the map transition between two geological ages to be animated.

1.4 Scope of represented groups

Clarification (MVP data scope). The MVP data set contains **non-avian dinosaurs only**; secondary Mesozoic reptile groups (FONC-380 / FONC-390) are introduced in V1. The main-vs-secondary distinction (FONC-360 / FONC-410) is therefore present in the MVP as labelling machinery applied to dinosaur content; it is not satisfied by shipping any secondary taxa in the MVP.

- **FONC-340** [MVP] — The system must treat non-avian dinosaurs as the main content.
- **FONC-350** [MVP] — The system must allow only non-avian dinosaurs to be displayed.

- **FONC-360** [MVP] — The system must distinguish non-avian dinosaurs from other included Mesozoic reptiles.
- **FONC-370** [MVP] — The system must cover the major dinosaur groups when data is available: theropods, sauropodomorphs, ornithomorphs, thyreophorans, ceratopsians and pachycephalosaurs.
- **FONC-380** [V1] — The system should include selected secondary groups among pterosaurs, marine reptiles, crocodylomorphs and Mesozoic lepidosaurs.
- **FONC-390** [V1] — The system should allow only included secondary groups to be displayed.
- **FONC-400** [MVP] — The system must not present the application as a complete atlas of all Mesozoic life.
- **FONC-410** [MVP] — The system must clearly indicate whether a taxon belongs to the main content or secondary content.
- **FONC-420** [V1] — The system should allow secondary groups to be hidden in order to preserve a dinosaur-centered experience.

1.5 Taxon content levels

- **FONC-430** [MVP] — The system must classify each displayed taxon according to a content level.
- **FONC-440** [MVP] — The system must support an Occurrence only level for taxa that only have basic fossil occurrence data.
- **FONC-450** [MVP] — The system must support a Short profile level for taxa that have a name, classification, time range and at least one occurrence.
- **FONC-460** [V1] — The system should support a Detailed profile level for taxa that have sourced morphological or ecological information.
- **FONC-470** [V1] — The system should support a Featured species level for species that have an enriched profile, a dedicated map and associated information.
- **FONC-480** [MVP] — The system must clearly indicate when a taxon profile is incomplete.
- **FONC-490** [MVP] — The system must display unavailable fields with an explicit label instead of an empty space.
- **FONC-500** [V1] — The system should allow taxa to be filtered or sorted by content level.

1.6 Taxon and species profile

- **FONC-510** [MVP] — The system must provide a profile for each taxon selectable from the map or search results.
- **FONC-520** [MVP] — The system must display the scientific name of the taxon on its profile.
- **FONC-530** [MVP] — The system must display the taxonomic rank of the taxon on its profile.
- **FONC-540** [MVP] — The system must display the available taxonomic classification of the taxon on its profile.
- **FONC-550** [MVP] — The system must display the known time range of the taxon on its profile.
- **FONC-560** [MVP] — The system must display the known fossil occurrences associated with the taxon.
- **FONC-570** [MVP] — The system must display the modern discovery locations associated with the taxon.
- **FONC-580** [MVP] — The system must display the reconstructed paleogeographic positions associated with the taxon.
- **FONC-590** [MVP] — The system must display the sources used for the main information on the profile.
- **FONC-600** [V1] — The system should display the assumed diet of the taxon when this information is available.

- **FONC-610 [V1]** — The system should display the assumed locomotion mode of the taxon when this information is available.
- **FONC-620 [V1]** — The system should display an estimated length when this information is available.
- **FONC-630 [V1]** — The system should display an estimated mass when this information is available.
- **FONC-640 [V1]** — The system should display a short descriptive summary for taxa at the Detailed profile or Featured species level.
- **FONC-650 [V1]** — The system should display related taxa when this information is available.
- **FONC-660 [V1]** — The system should display taxa known from the same formation or region and from an overlapping time range.
- **FONC-670 [MVP]** — The system must indicate when profile information is interpretative rather than directly derived from a fossil occurrence.

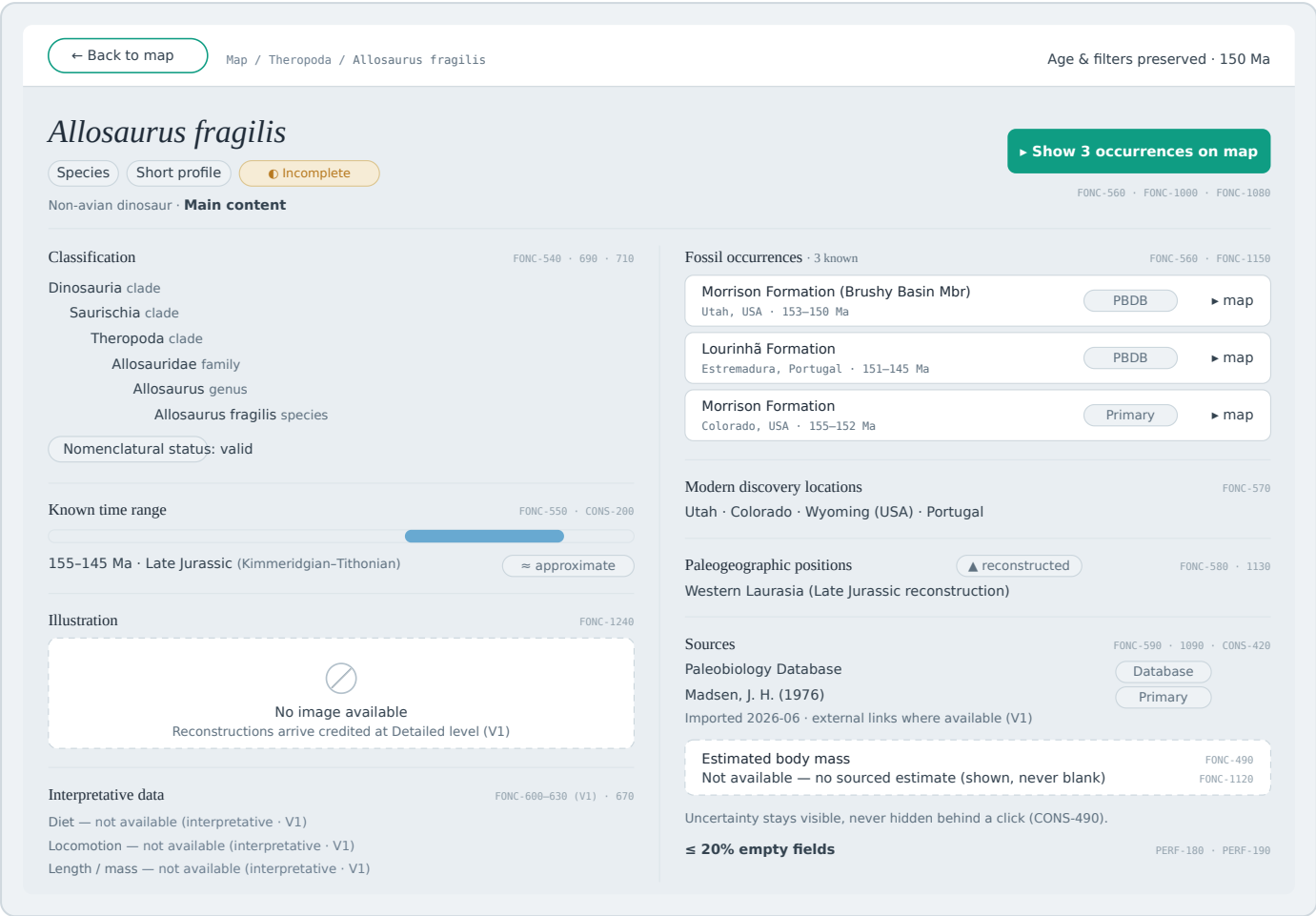


Figure 4 — UI mockup: Taxon profile

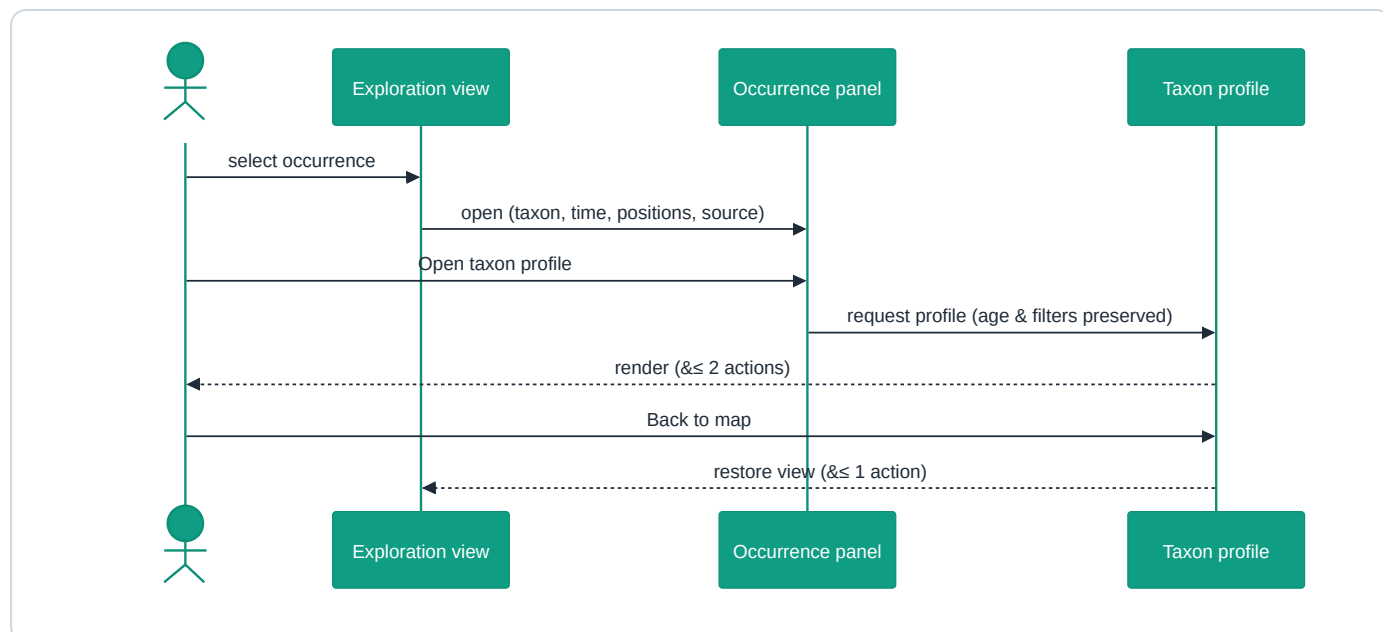


Figure 5 — Sequence — occurrence to profile and back

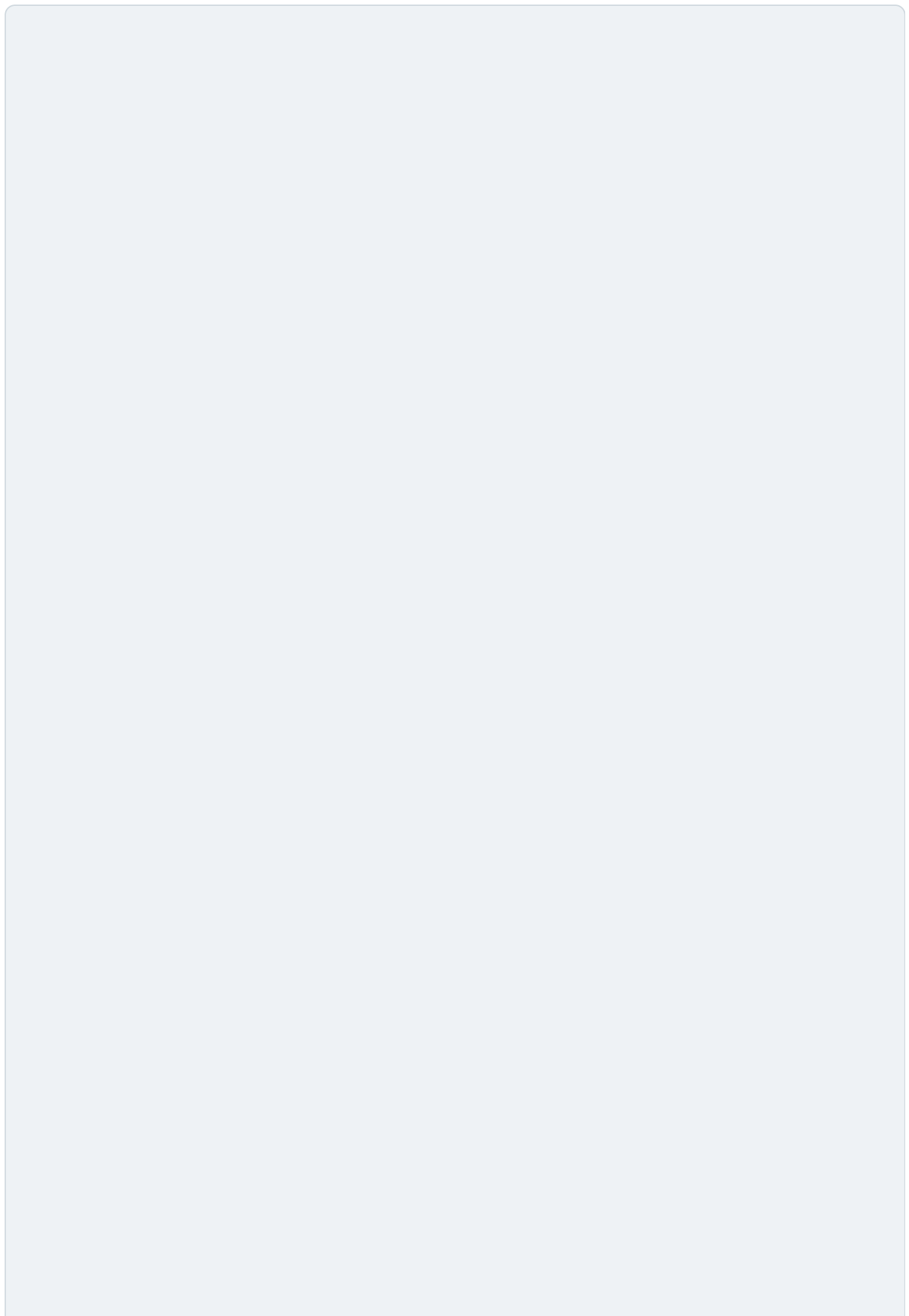
1.7 Classification and taxonomy

- **FONC-680** [MVP] — The system must allow taxa to be explored by taxonomic group.
- **FONC-690** [MVP] — The system must display the available taxonomic hierarchy of a selected taxon.
- **FONC-700** [MVP] — The system must allow a taxonomic group to be selected and the map to be updated according to that group.
- **FONC-710** [MVP] — The system must distinguish the displayed taxonomic levels, for example clade, family, genus and species.
- **FONC-720** [MVP] — The system must indicate when a taxon is invalid, doubtful, synonymous or uncertain when this information is available.
- **FONC-730** [V1] — The system should allow navigation from a species to its parent groups.
- **FONC-740** [V1] — The system should allow navigation from a group to its associated genera or species.
- **FONC-750** [V1] — The system should allow the map to be filtered from a selection in the taxonomic tree.

1.8 Search and filters

- **FONC-760** [MVP] — The system must allow a taxon to be searched by scientific name.
- **FONC-770** [V1] — The system should allow a taxon to be searched by common name or popular name when one is available.
- **FONC-780** [MVP] — The system must allow occurrences to be filtered by geological period.
- **FONC-790** [MVP] — The system must allow occurrences to be filtered by taxonomic group.
- **FONC-800** [MVP] — The system must allow occurrences to be filtered to display only non-avian dinosaurs.
- **FONC-810** [V1] — The system should allow occurrences to be filtered by diet.
- **FONC-820** [V1] — The system should allow occurrences to be filtered by modern discovery region.
- **FONC-830** [V1] — The system should allow occurrences to be filtered by taxon content level.
- **FONC-840** [V1] — The system should allow occurrences to be filtered by data completeness or reliability level.
- **FONC-850** [MVP] — The system must display the number of results matching the active filters.

- **FONC-860** [MVP] — The system must display an explicit empty state when no result matches the active filters.
- **FONC-870** [MVP] — The system must allow an active filter to be removed individually.
- **FONC-880** [MVP] — The system must allow all active filters to be removed in a single action.



Filters

3 active

PERIOD

FONC-780

- ☐ Triassic 252-201 Ma
- ☒ **Jurassic** 201-145 Ma
- ☐ Cretaceous 145-66 Ma

TAXONOMIC GROUP

FONC-790

- ☒ **Theropoda** 47
- ☐ Sauropodomorpha 31
- ☐ Ornithopoda 12
- ☐ Thyreophora (Stegosauria) 9
- ☐ Ceratopsia Cretaceous only
- ☐ Pachycephalosauria Cretaceous only

2 groups have no occurrences at 150 Ma.

Dinosaurs only



Hides secondary Mesozoic reptiles (V1).

ACTIVE FILTERS

FONC-870

- Jurassic** ×
- Theropoda** ×
- Dinosaurs only** ×

47 occurrences match
of 99 Theropoda across the Jurassic FONC-060 · 850

↺ Reset all filters

Diet · region · completeness filters — V1

FONC-880

F0NC-800 · F0NC-020 · states: active / disabled

Figure 6 — UI mockup: Filters panel

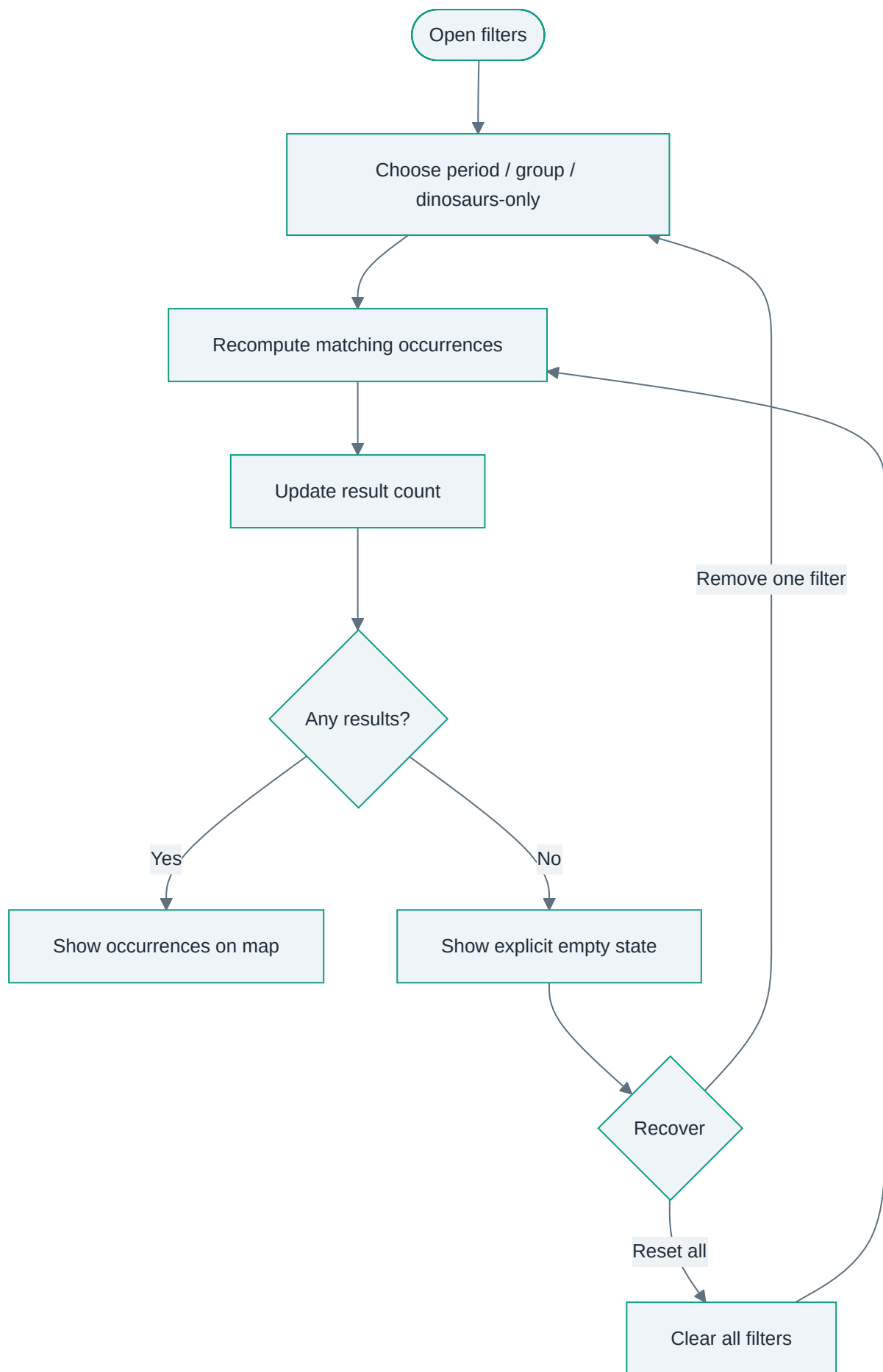


Figure 7 — Activity — filter application & empty state

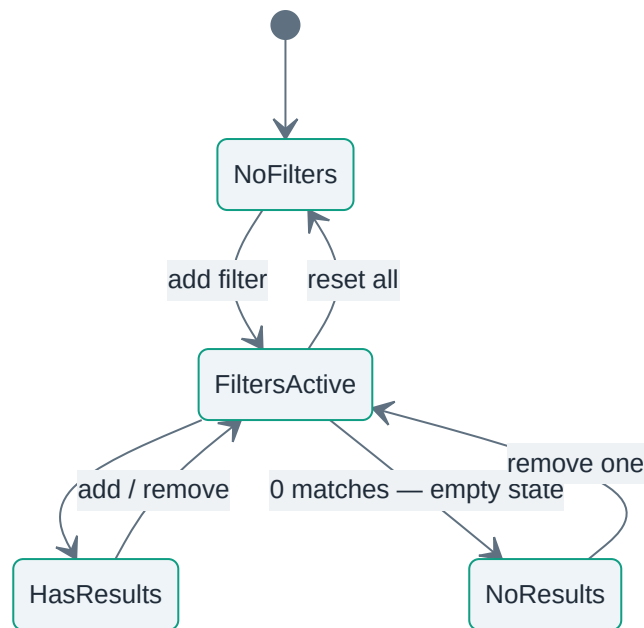


Figure 8 — State — active filter set

1.9 Locations, occurrences and formations

- **FONC-890** [MVP] — The system must provide a profile or information panel for each selected fossil occurrence.
- **FONC-900** [MVP] — The system must display the modern location associated with a fossil occurrence when this information is available.
- **FONC-910** [MVP] — The system must display the reconstructed paleogeographic position associated with a fossil occurrence when this information is available.
- **FONC-920** [MVP] — The system must display the time range associated with a fossil occurrence.
- **FONC-930** [MVP] — The system must display the source associated with a fossil occurrence.
- **FONC-940** [V1] — The system should display the name of the geological formation associated with an occurrence when this information is available.
- **FONC-950** [V1] — The system should provide a profile for a selected geological formation.
- **FONC-960** [V1] — The system should display the taxa known from a selected geological formation.
- **FONC-970** [V1] — The system should display the age or time range of a geological formation when this information is available.
- **FONC-980** [V1] — The system should allow comparison between the modern position and paleogeographic position of a selected location.

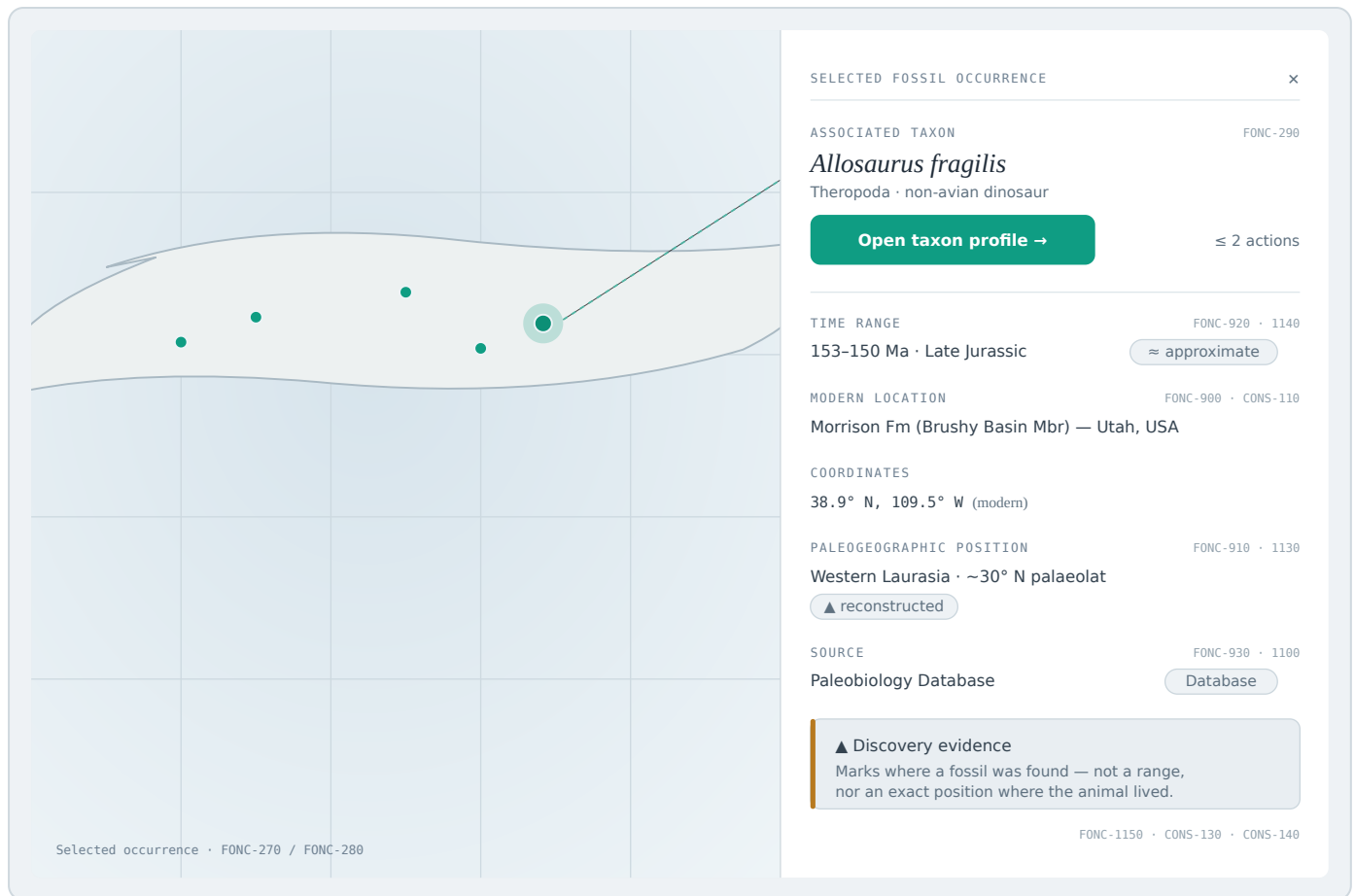


Figure 9 — UI mockup: Occurrence panel

1.10 Navigation between content

- **FONC-990** [MVP] — The system must allow a taxon profile to be opened from a selected fossil occurrence.
- **FONC-1000** [MVP] — The system must allow the user to return to the map from a taxon profile.
- **FONC-1010** [MVP] — The system must preserve the selected age when navigating from the map to a taxon profile.
- **FONC-1020** [MVP] — The system must preserve the active filters when navigating from the map to a taxon profile.
- **FONC-1030** [V1] — The system should allow a formation profile to be opened from a selected fossil occurrence.
- **FONC-1040** [V1] — The system should allow a taxon profile to be opened from a formation profile.
- **FONC-1050** [V1] — The system should allow a related taxon profile to be opened from a taxon profile.
- **FONC-1060** [V1] — The system should allow the user to return to the previous search result after viewing a profile.
- **FONC-1070** [MVP] — The system must allow access to a taxon profile in a maximum of 2 actions from an occurrence visible on the map.
- **FONC-1080** [MVP] — The system must allow the user to return to the map in a maximum of 1 action from a taxon profile.

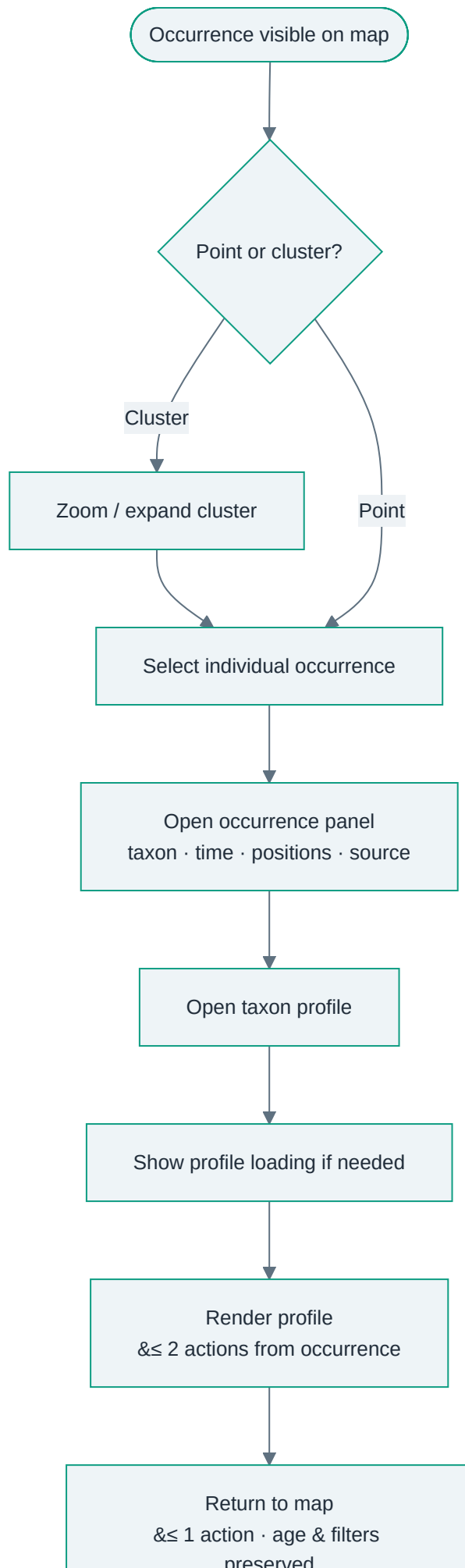


Figure 10 — Activity — occurrence selection to taxon profile

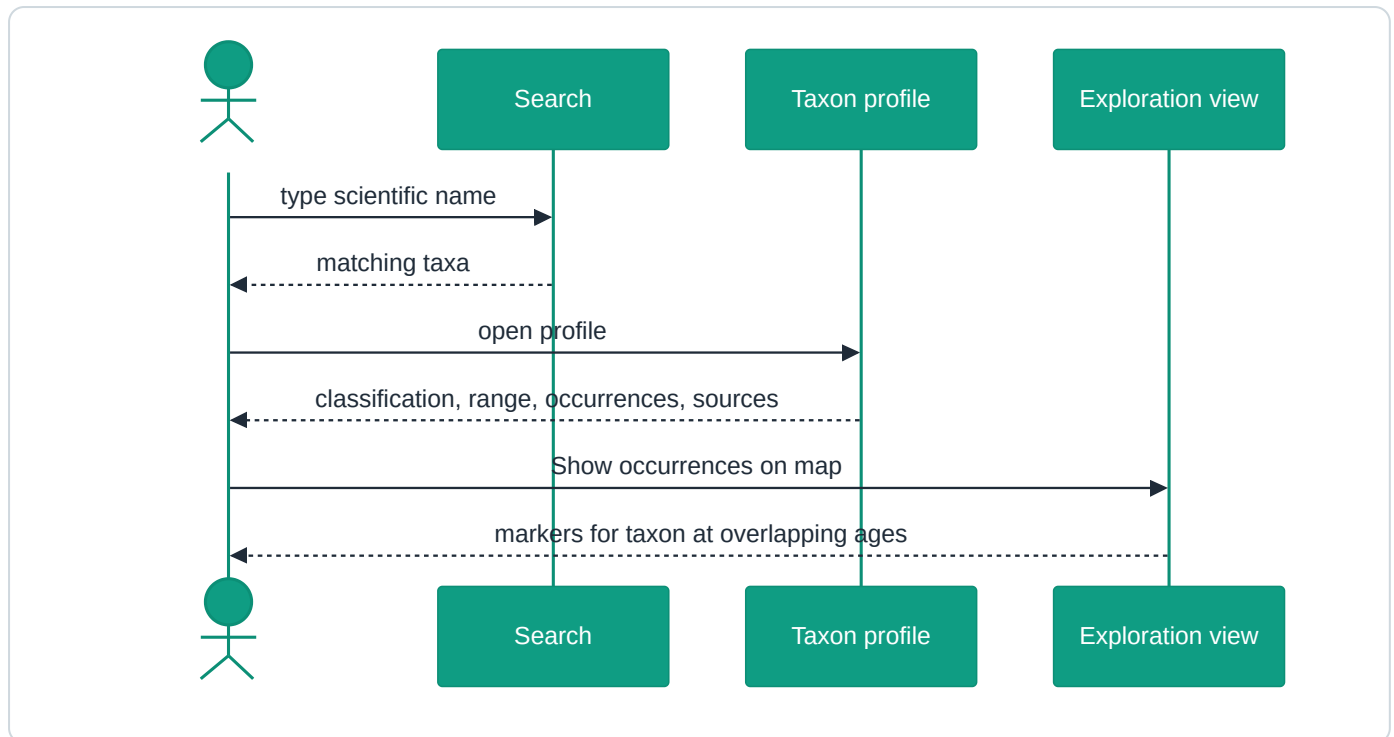


Figure 11 — Sequence — search to occurrences on map

1.11 Sources, provenance and uncertainty

Clarification (which sources). These requirements demand an *identifiable* source for every visible occurrence and time range but deliberately do **not** mandate a particular provider. The concrete dataset(s) are a technical-design decision recorded in the [data architecture design](#) (Paleobiology Database + Wikipedia/Wikidata); any provider that supplies sourced occurrences, time ranges and reconstructions satisfies the requirements below.

- **FONC-1090** [MVP] — The system must display the sources of the scientific data used.
- **FONC-1100** [MVP] — The system must display an identifiable source for each visible fossil occurrence.
- **FONC-1110** [MVP] — The system must distinguish data directly derived from fossil occurrences from interpretative data.
- **FONC-1120** [MVP] — The system must indicate when information is missing instead of replacing it with an unmarked assumption.
- **FONC-1130** [MVP] — The system must indicate when a geographic position is reconstructed.
- **FONC-1140** [MVP] — The system must indicate when a time range is approximate.
- **FONC-1150** [MVP] — The system must display fossil occurrences as evidence of discovery, not as complete distribution boundaries.
- **FONC-1160** [V1] — The system should display a confidence level for morphological, ecological or behavioral information when such a level is available.
- **FONC-1170** [V1] — The system should display the consultation or import date of the data when this information is available.
- **FONC-1180** [V1] — The system should display a link to the external source when this link is available.

1.12 Images and illustrations

- **FONC-1190** [V1] — The system should display an image or illustration for taxa at the Detailed profile or Featured species level when this resource is available.
- **FONC-1200** [V1] — The system must distinguish between a fossil photograph, an artistic reconstruction and a silhouette when an image is displayed.
- **FONC-1210** [V1] — The system must display the source or credit for an image when the image is displayed.
- **FONC-1220** [V1] — The system must indicate when an image is an artistic representation.
- **FONC-1230** [V1] — The system must not present an artistic illustration as direct evidence of the animal's exact appearance.
- **FONC-1240** [MVP] — The system must display an alternative state when no image is available for a taxon.
- **FONC-1250** [V2] — The system should allow the animal's size to be visually compared with a human reference when dimensions are available.

1.13 Interface states

- **FONC-1260** [MVP] — The system must display a loading state while the map initially loads.
- **FONC-1270** [MVP] — The system must display a loading state when opening a taxon profile if the data is not already available.
- **FONC-1280** [MVP] — The system must display an empty state when the active filters return no occurrence.
- **FONC-1290** [MVP] — The system must display an empty state when a search returns no taxon.
- **FONC-1300** [MVP] — The system must display an explicit message when a taxon profile exists but contains only minimal data.
- **FONC-1310** [MVP] — The system must display a clear error message when a map cannot be loaded.
- **FONC-1320** [MVP] — The system must display a clear error message when a taxon profile cannot be loaded.
- **FONC-1330** [MVP] — The system must allow a failed load to be retried.
- **FONC-1340** [MVP] — The system must preserve active filters after a profile, map or occurrence loading failure.

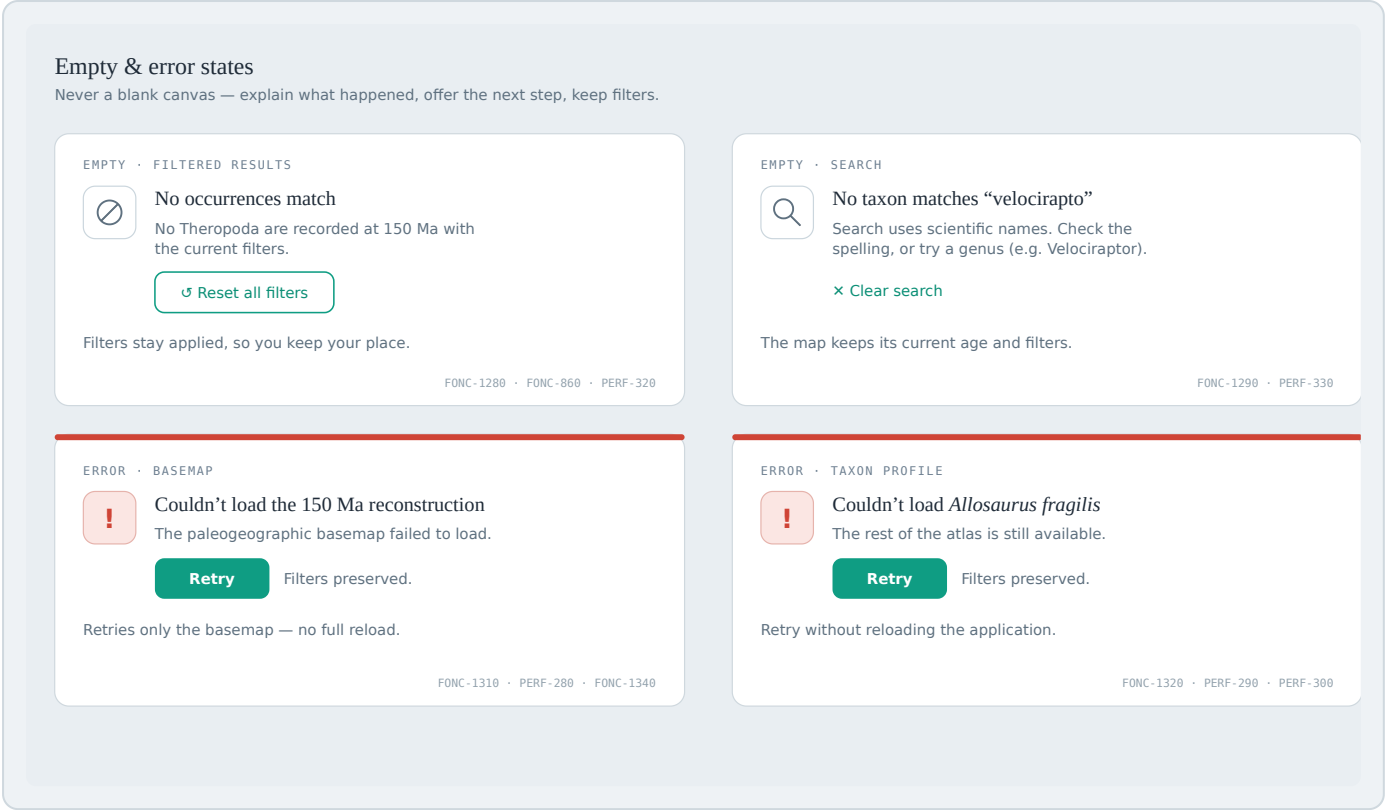


Figure 12 — UI mockup: Empty & error states

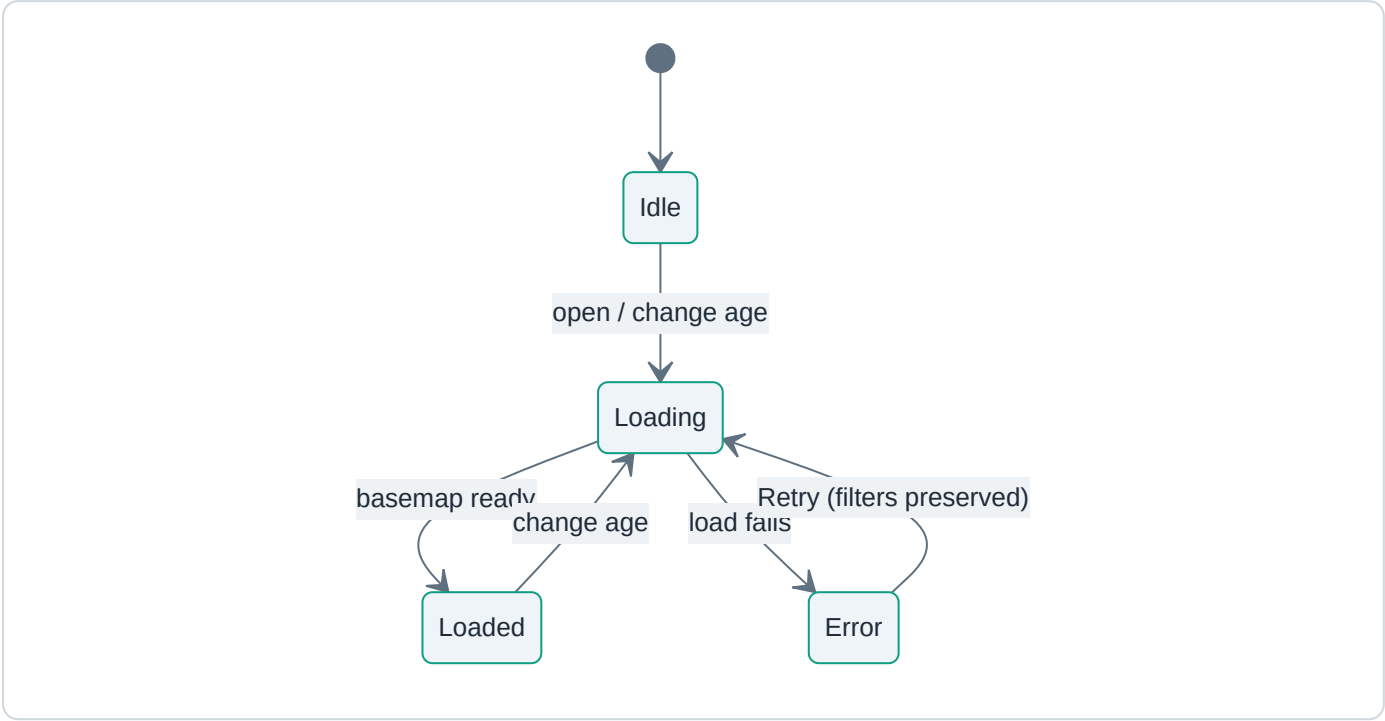


Figure 13 — State — map / basemap loading

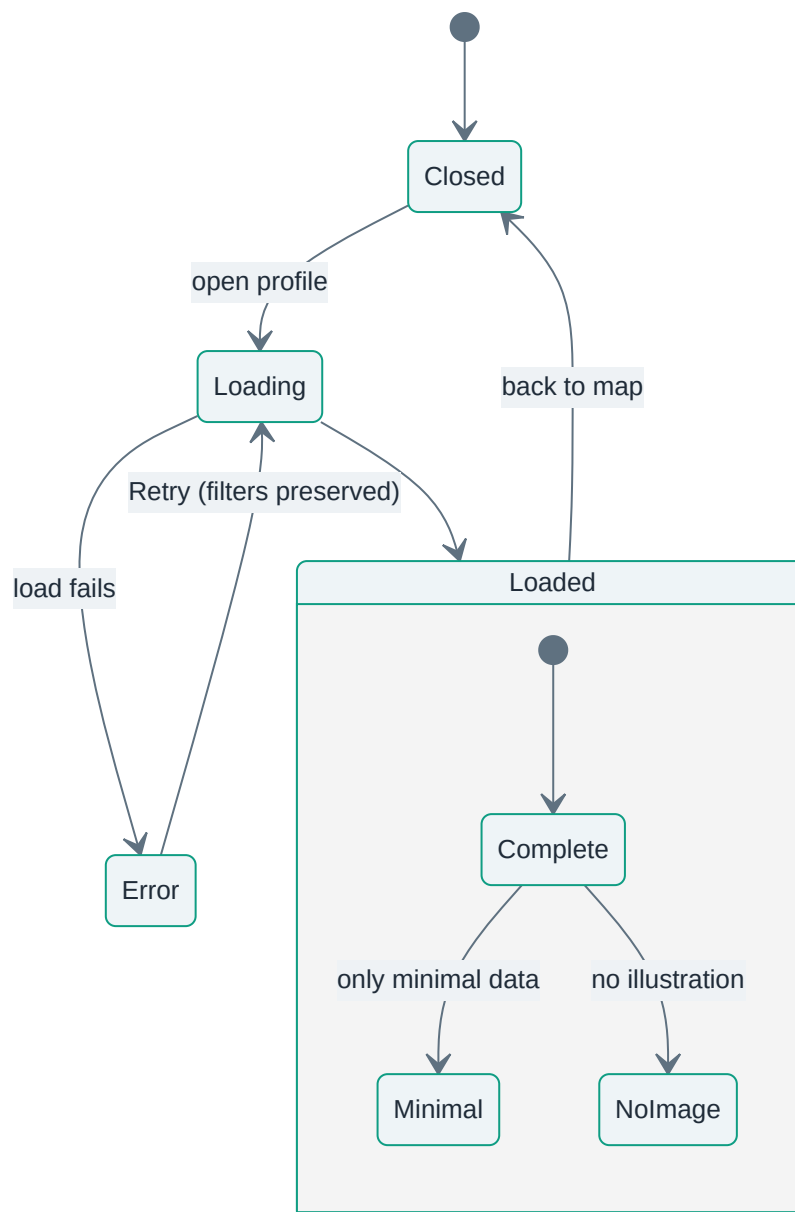


Figure 14 — State — taxon profile loading

1.14 Comparison and guided paths

- **FONC-1350** [V2] — The system should allow two geological periods to be compared in a dedicated view.
- **FONC-1360** [V2] — The system should allow two taxonomic groups to be compared in a dedicated view.
- **FONC-1370** [V1] — The system should display a timeline for a selected taxon.
- **FONC-1380** [V1] — The system should display the known first and last appearance of a taxon on a timeline.
- **FONC-1390** [V2] — The system should allow a region to be selected and display the taxa known from that region for the selected age.
- **FONC-1400** [V2] — The system should provide thematic guided paths.
- **FONC-1410** [V2] — The system should provide at least one guided path focused on large theropods.
- **FONC-1420** [V2] — The system should provide at least one guided path focused on Jurassic dinosaurs.

- **FONC-1430** [V2] — The system should provide at least one guided path focused on Late Cretaceous dinosaurs.
-

2. Constraints

2.1 Scientific scope

- **CONS-010** [MVP] — The system must limit its main temporal scope to the Mesozoic, between 252 Ma and 66 Ma.
- **CONS-020** [MVP] — The system must exclude species after the Cretaceous–Paleogene extinction, except in explanatory content explicitly marked as context.
- **CONS-030** [MVP] — The system must treat non-avian dinosaurs as the main scope.
- **CONS-040** [MVP] — The system must treat other Mesozoic reptiles as secondary scope.
- **CONS-050** [MVP] — The system must not cover all Mesozoic plants, invertebrates, mammals, fish or microorganisms in its main scope.
- **CONS-060** [MVP] — The system must display only taxa with at least one identifiable source.
- **CONS-070** [MVP] — The system must display only occurrences with at least one identifiable source.
- **CONS-080** [MVP] — The system must not display a numeric value for size, mass or age without a source or uncertainty indication.
- **CONS-090** [V1] — The system should limit secondary groups to a list explicitly defined in the interface or product documentation.
- **CONS-100** [MVP] — The system must avoid wording that implies secondary content has the same editorial status as dinosaurs.

2.2 Geographic representation

- **CONS-110** [MVP] — The system must distinguish modern coordinates from reconstructed paleogeographic coordinates.
- **CONS-120** [MVP] — The system must state that ancient maps are scientific reconstructions and not direct observations.
- **CONS-130** [MVP] — The system must not present fossil points as exact positions where the animal lived.
- **CONS-140** [MVP] — The system must present fossil points as documented discovery, collection or fossil observation locations.
- **CONS-150** [MVP] — The system must not automatically extrapolate a continuous distribution area from isolated fossil points.
- **CONS-160** [MVP] — The system must not display a complete distribution area unless that area is explicitly sourced.
- **CONS-170** [V1] — The system should display an uncertainty note when the geographic resolution of an occurrence is insufficient for the displayed map scale.
- **CONS-180** [MVP] — The system must use consistent geographic units throughout the interface.

2.3 Temporal representation

- **CONS-190** [MVP] — The system must represent geological ages in millions of years before present.

- **CONS-200** [MVP] — The system must display time ranges with a minimum boundary and a maximum boundary when both values are available.
- **CONS-210** [MVP] — The system must clearly indicate when a species is known from a broad time range.
- **CONS-220** [MVP] — The system must not present a species as present at a precise age when only a time range is available.
- **CONS-230** [MVP] — The system must not display two taxa as contemporaneous if their known time ranges do not overlap.
- **CONS-240** [V1] — The system should indicate the period, epoch or geological stage associated with a time range when this information is available.

2.4 Scientific consistency

- **CONS-250** [MVP] — The system must not display a taxon outside its known time range.
- **CONS-260** [MVP] — The system must not display an occurrence outside its known time range.
- **CONS-270** [MVP] — The system must not present a predator-prey relationship as certain without an explicit source.
- **CONS-280** [MVP] — The system must not present assumed behavior as an established fact.
- **CONS-290** [MVP] — The system must distinguish fossil facts, scientific estimates and hypotheses.
- **CONS-300** [MVP] — The system must indicate when a taxon is controversial, invalid, doubtful or synonymous when this information is available.
- **CONS-310** [V1] — The system should display an explanatory note when a taxon known to the public is not recognized as valid in the data used.

2.5 Editorial content

- **CONS-320** [MVP] — The system must use an informative, scientific and accessible tone.
- **CONS-330** [MVP] — The system must not use unverifiable sensationalist wording.
- **CONS-340** [MVP] — The system must avoid absolute claims when data is incomplete or uncertain.
- **CONS-350** [MVP] — The system must use scientific names as the primary reference for taxa.
- **CONS-360** [V1] — The system should use common names as reading aids when they are available.
- **CONS-370** [MVP] — The system must display units for numeric values, for example meters, kilograms or millions of years.
- **CONS-380** [V1] — The system should avoid an overly childish art direction in order to preserve a scientific atlas identity.

2.6 Sources and provenance

- **CONS-390** [MVP] — The system must associate each visible fossil occurrence with an identifiable source.
- **CONS-400** [MVP] — The system must associate each displayed time range with an identifiable source or dataset.
- **CONS-410** [MVP] — The system must associate each size or mass estimate with a source or uncertainty indication.
- **CONS-420** [MVP] — The system must distinguish a primary source, a database and an editorial synthesis when this information is available.
- **CONS-430** [V1] — The system should display the consultation or import date of an external source.
- **CONS-440** [MVP] — The system must not mix sourced data and unmarked assumptions in the same field.

2.7 Interface and usability

- **CONS-450** [MVP] — The system must keep the main time, map and filter controls visible on the exploration view.
 - **CONS-460** [MVP] — The system must allow access to a taxon profile in a maximum of 2 actions from an occurrence visible on the map.
 - **CONS-470** [MVP] — The system must allow the user to return to the map from a taxon profile in a maximum of 1 action.
 - **CONS-480** [MVP] — The system must maintain consistent nomenclature for periods, groups and taxa throughout the interface.
 - **CONS-490** [MVP] — The system must avoid hiding uncertainty information behind a secondary interaction when that information changes the scientific interpretation of the content.
 - **CONS-500** [V1] — The system should allow the application to be used on desktop and tablet screens.
 - **CONS-510** [V2] — The system should allow the application to be used on mobile phones without loss of the main features.
-

3. Performance

3.1 Response time

- **PERF-010** [MVP] — The system must display the main view in 5 seconds maximum after the initial load on a standard desktop connection.
- **PERF-020** [MVP] — The system must display the first useful content in 3 seconds maximum after the initial load.
- **PERF-030** [MVP] — The system must update visible occurrences in 1 second maximum after an age change when the required data is already loaded.
- **PERF-040** [MVP] — The system must open a taxon profile in 1 second maximum when the profile data is already loaded.
- **PERF-050** [MVP] — The system must display a loading indicator when a view update exceeds 500 milliseconds.
- **PERF-060** [MVP] — The system must provide visual feedback in 100 milliseconds maximum after a zoom or pan interaction on the map.
- **PERF-070** [V1] — The system should perform a simple text search in 500 milliseconds maximum when the required data is already loaded.

3.2 Readability and map density

- **PERF-080** [MVP] — The system must keep main labels readable at a minimum size of 12 CSS px.
- **PERF-090** [MVP] — The system must prevent more than 30 individual markers from visually overlapping in a 100 × 100 CSS px area without grouping.
- **PERF-100** [MVP] — The system must group occurrences into clusters when their density prevents reliable selection of individual points.
- **PERF-110** [MVP] — The system must display at least 3 zoom levels: global, regional and local.
- **PERF-120** [MVP] — The system must ensure that each displayed point or cluster is selectable with a minimum interactive area of 24 × 24 CSS px.

- **PERF-130** [V1] — The system should display taxon or formation names only when the zoom level allows them to be read without excessive overlap.

3.3 Minimum quality of visible data

- **PERF-140** [MVP] — The system must display an identifiable source for 100% of visible fossil occurrences.
- **PERF-150** [MVP] — The system must display a time range for 100% of visible fossil occurrences.
- **PERF-160** [MVP] — The system must display a minimum taxonomic classification for 100% of visible taxa.
- **PERF-170** [MVP] — The system must display at least one parent group above genus or species for 100% of visible taxa when this information is available.
- **PERF-180** [MVP] — The system must explicitly signal missing fields in 100% of affected taxon profiles.
- **PERF-190** [MVP] — The system must not display a taxon profile containing more than 20% silently empty fields.
- **PERF-200** [V1] — The system should display at least 50 detailed dinosaur profiles in the first enriched version.
- **PERF-210** [V1] — The system should display at least 10 featured species in the first enriched version.

3.4 Accessibility and usability

- **PERF-220** [MVP] — The system must allow the main view to be used with a mouse or trackpad.
- **PERF-230** [MVP] — The system must allow the main filters to be used with a keyboard.
- **PERF-240** [MVP] — The system must maintain sufficient contrast for main text to remain readable.
- **PERF-250** [MVP] — The system must display critical information without relying only on color.
- **PERF-260** [V1] — The system should provide alternative or descriptive text for species images when they are displayed.
- **PERF-270** [MVP] — The system must allow a point or cluster to be selected without requiring click precision below 24 × 24 CSS px.

3.5 Display robustness

- **PERF-280** [MVP] — The system must display a clear error message when map data cannot be loaded.
- **PERF-290** [MVP] — The system must display a clear error message when a taxon profile cannot be loaded.
- **PERF-300** [MVP] — The system must allow a failed load to be retried without reloading the entire application.
- **PERF-310** [MVP] — The system must preserve active filters after the loading failure of a profile, map or occurrence.
- **PERF-320** [MVP] — The system must display an empty state when active filters return no occurrence.
- **PERF-330** [MVP] — The system must display an empty state when search returns no result.

3.6 MVP functional validation

- **PERF-340** [MVP] — The system must allow the full scenario "select a period → filter dinosaurs → select an occurrence → open a taxon profile → return to the map" to be completed without a blocking error.
- **PERF-350** [MVP] — The system must allow the full scenario "search for a taxon → open its profile → display its occurrences on the map" to be completed without a blocking error.

- **PERF-360** [MVP] — The system must allow the full scenario "change the selected age → observe the change in visible occurrences" to be completed without a full application reload.
 - **PERF-370** [MVP] — The system must allow the full scenario "activate a filter with no result → display an empty state → reset filters" to be completed without a blocking error.
-

4. MVP Scope Summary

The MVP must allow users to:

1. Explore the Mesozoic between 252 Ma and 66 Ma.
2. Display a paleogeographic map corresponding to the selected age.
3. Display fossil occurrences of non-avian dinosaurs.
4. Filter occurrences by period and taxonomic group.
5. Select a fossil occurrence.
6. Open a taxon profile from an occurrence.
7. Display classification, time range, discovery locations and sources.
8. Distinguish direct fossil data from interpretative data.
9. Clearly display missing, uncertain or reconstructed data.
10. Avoid misleading representation of actual species distribution.

5. Explicitly Out of MVP Scope

The MVP does not need to include:

1. Continuous animation of continental drift.
2. A 3D globe view.
3. Side-by-side comparison between two periods.
4. A complete phylogeny.
5. Guided paths.
6. Complete coverage of all Mesozoic reptiles.
7. Complete coverage of all Mesozoic life.
8. Artistic reconstructions for all taxa.
9. Systematic predator-prey relationships.
10. Extrapolated distribution range maps.

The map is a 2-D paleogeographic view. The **3D globe view (item 2)** is an explicit product non-goal, not merely a deferred feature: adding it would require a new (V2) requirement. The remaining items are deferred to V1 or V2 as indicated by the priorities in §§1–3.

Data architecture & model (design)

Status: Technical design document. This is **not** a source of product requirements — those live only in the **functional specification**. This document records *how* the data behind the atlas is sourced, structured, and served so that the requirements (especially the provenance and uncertainty rules of §1.11 and §2.x) are satisfied **structurally** rather than by convention.

It fixes decisions the functional specification deliberately leaves open (e.g. which datasets supply the data) and is the reference for the class models of the data layer.

1. Principles

Three ideas drive everything below.

1. **The atlas is a curation & presentation layer, not a source of truth for paleobiology.** We do not re-derive taxonomy, dating, or plate reconstructions. Being a data *authority* is the single largest project risk; we avoid it by leaning on established datasets and adding only a thin editorial layer.
2. **Separate facts from assertions from derivations.** A datum is one of: - an **imported fact** (an occurrence record, a measurement) — sourced; - an **assertion** *about* something (validity, diet, mass) — a claim made *by a source, at a date*, not an intrinsic property; or - a **derived value** (accepted taxonomy, "approximate", "reconstructed", content level) — a deterministic function of the two above. Most of the enums that look like intrinsic attributes are really assertions or derivations. Modelling them that way is what makes provenance honest.
3. **Provenance is a structural invariant.** Every value the UI can display resolves to a Source (or is explicitly marked *Editorial*). It is not possible to render a field without a provenance pointer.

2. Sources and their roles

Two external datasets, joined through **Wikidata**.

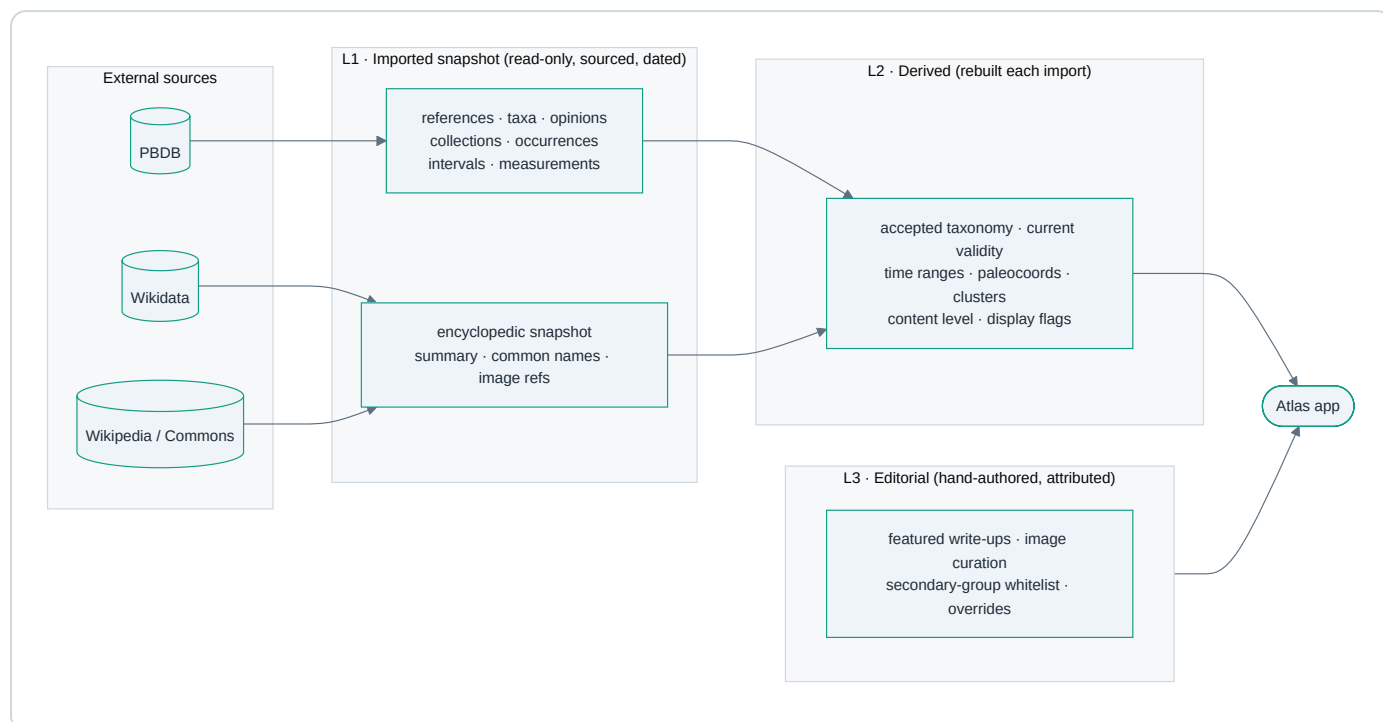
Source	Role	Provenance kind	Licence
Paleobiology Database (PBDB)	Scientific spine — occurrences, collections, taxa, taxonomic opinions , references, geologic intervals, measurements, and paleocoordinates	Database → resolves to PrimaryLiterature (PBDB records the citing reference)	CC0 / CC BY (per PBDB terms)
Wikipedia / Wikimedia Commons	Encyclopedic layer — narrative summary, common names, illustrations	Encyclopedic (tertiary)	Text CC BY-SA 4.0; images per-file (varies)
Wikidata	The join key , not display content — carries the PBDB taxon id, the Wikipedia sitelink, and the Commons image (P18) for a taxon	—	CC0

Why Wikidata is the bridge. Matching PBDB taxa to Wikipedia articles by name is fragile (homonyms, synonyms, spelling). A Wikidata item for a taxon typically holds its scientific name, a *Paleobiology Database taxon ID*, a sitelink to the Wikipedia article, common names, and a Commons image. Joining **by Wikidata QID** turns reconciliation into a lookup. Taxa without a Wikidata match simply have no encyclopedic content → the UI shows the explicit "not available" state (on brand).

Trust boundary. Wikipedia content is **tertiary** and always labelled as such (`SourceKind = Encyclopedic`). It must never be presented with the same editorial status as PBDB/primary data (CONS-100, CONS-330, CONS-340). Where Wikipedia states a numeric value (mass, length) we surface it marked tertiary; we never invent a confidence it did not state (CONS-080, CONS-410), and we prefer the primary reference Wikipedia itself cites when we can resolve it.

Licensing is a hard constraint. Displaying a Wikipedia summary requires attribution and share-alike (CC BY-SA 4.0). Every Commons image stores its own `licence`, `credit`, and `sourceUrl`, and is shown with them (FONC-1210); artistic reconstructions are labelled as such (FONC-1220/1230). An image whose licence we cannot honour is simply not shown → image-fallback state (FONC-1240).

3. Three data tiers

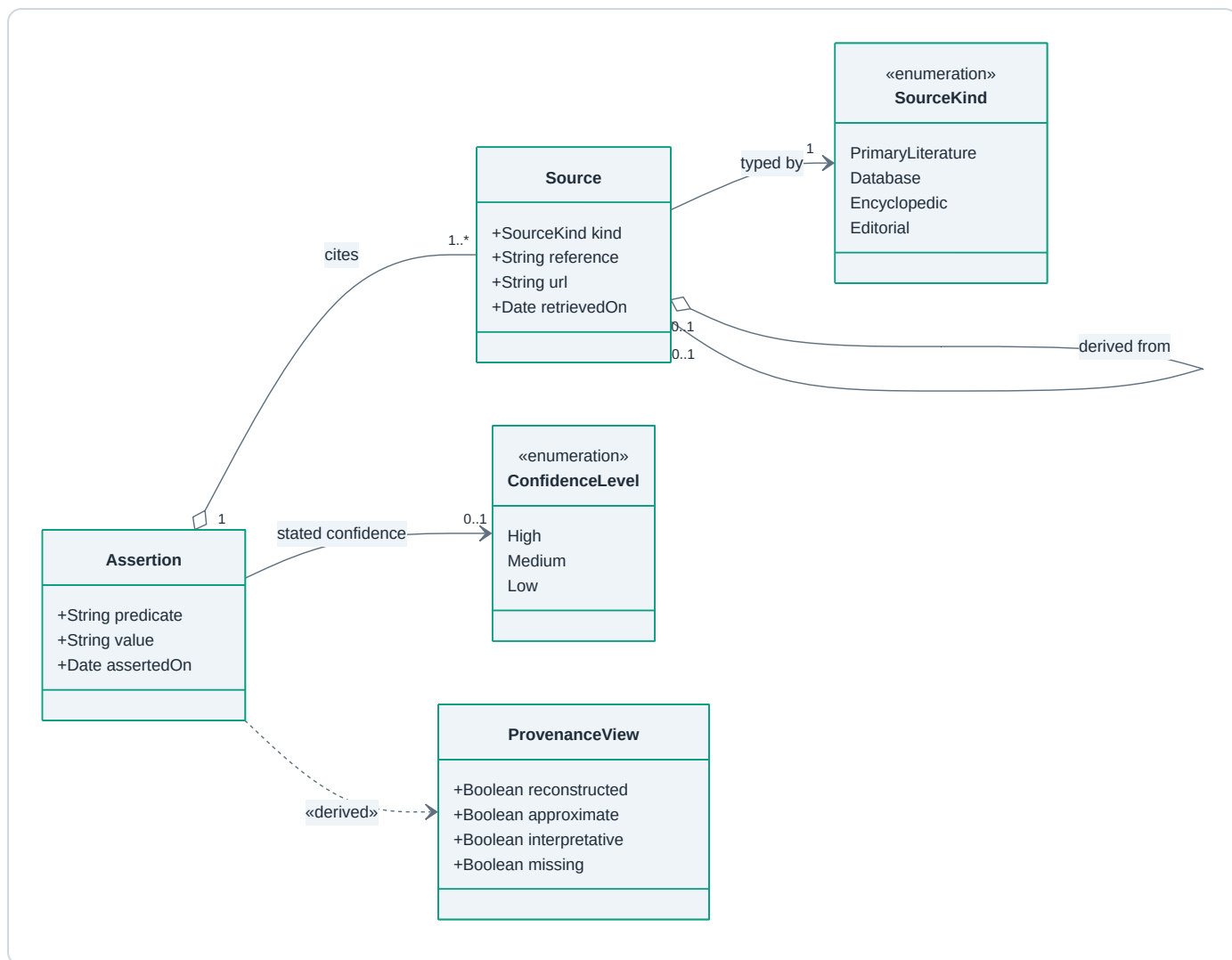


- **L1 — Imported snapshot.** Immutable, dated mirror of a PBDB subset plus an encyclopedic snapshot. Each row keeps its `Source` and the snapshot's `retrievedOn` date — this is where FONC-1170 / CONS-430 (import date) come from.
- **L2 — Derived.** A pure function of L1, rebuilt on every import: accepted taxonomy, current validity (from opinions), per-taxon time range, paleocoords, spatial clusters, `ContentLevel`, and the derived display flags. Never hand-edited.
- **L3 — Editorial.** Small, versioned, clearly attributed as *Editorial synthesis*: featured-species prose, image selection, the secondary-group whitelist (CONS-090), and rare curator overrides.

The app reads **L2 + L3 only**, at runtime, from our own store — it never calls PBDB or Wikipedia live (see §6). That keeps performance (PERF-010...040), offline reproducibility, and the import-date guarantee intact.

4. Provenance & the assertion pattern

The heart of the model. A claim is a first-class object with a source and a date; "validity", "diet", "mass" are all the same shape. The UI shows the **winning** assertion and its citation.



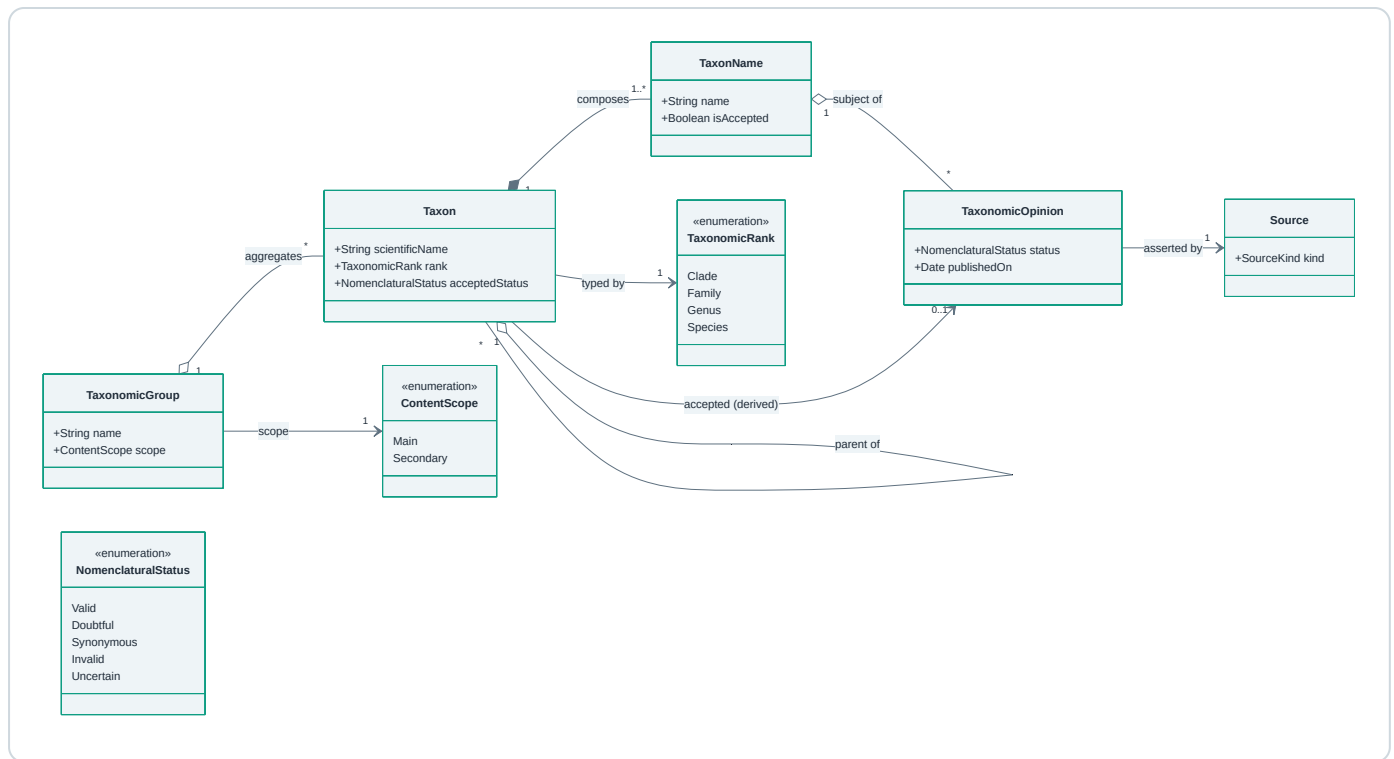
- **Source.derivedFrom** models the chain: an Editorial note cites an Encyclopedic article; a Database (PBDB) record aggregates a PrimaryLiterature reference. One flat enum could never express this.
- **ConfidenceLevel** is optional and only ever *stated by a source*. We do not compute or invent it.
- **ProvenanceView** is *«derived»*, not stored — the dashed dependency is the point. The four flags are computed from structure, so they can never drift from the data:

Display flag	Derived from
reconstructed	the value is a paleocoordinate (produced by a rotation model) — true by construction
approximate	the TimeRange spans more than one geologic stage, or has wide bounds
interpretative	the value's Source.kind is Encyclopedic or Editorial (not Database/PrimaryLiterature)
missing	the value is null

This directly answers "how would you set that value from sources": you don't set these — you compute them.

5. Taxonomy & validity (opinion-based)

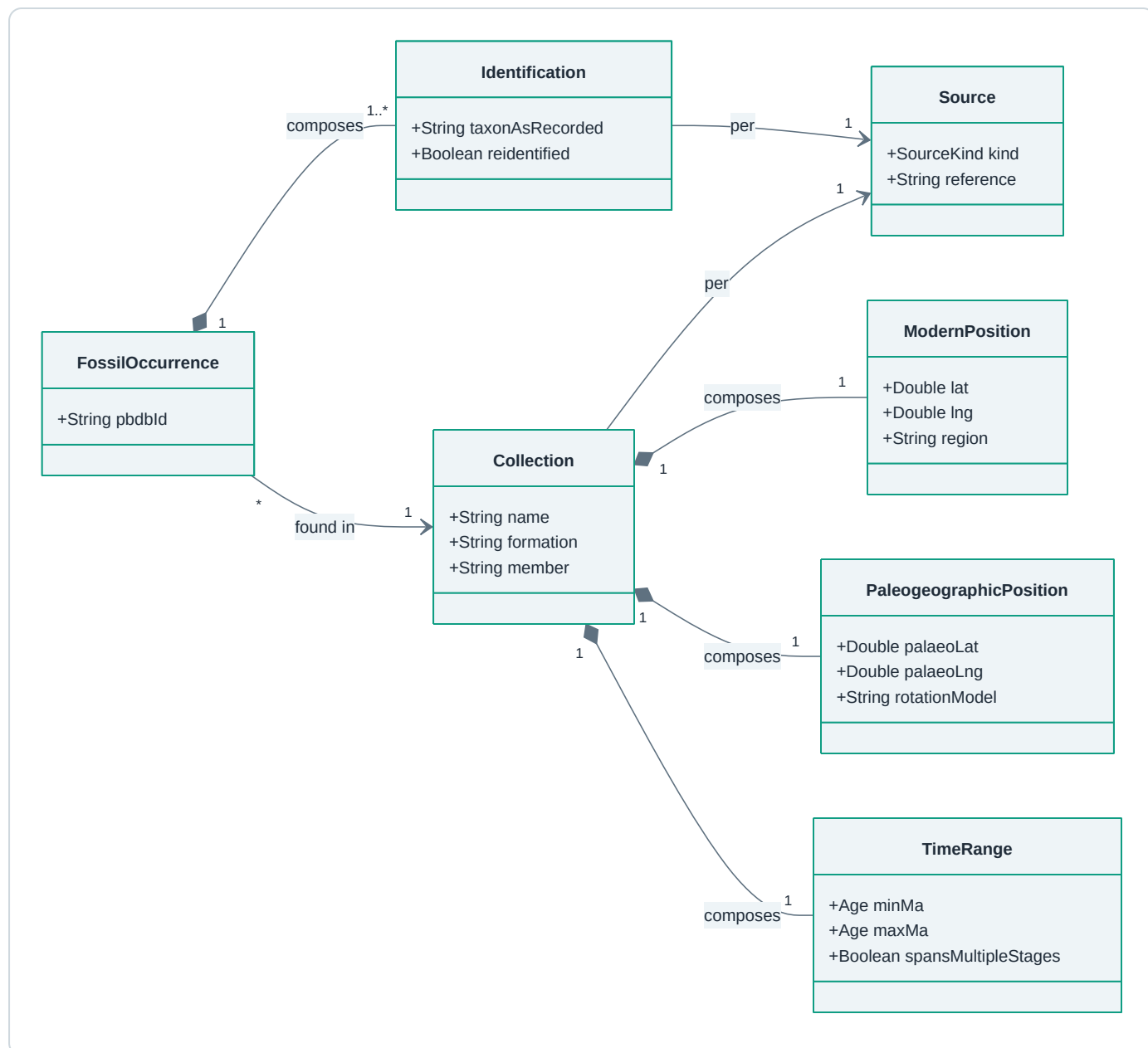
ValidityStatus is **not** an attribute of a taxon; it is derived from the set of **taxonomic opinions**, each asserted by a reference at a date. This mirrors how PBDB itself computes accepted names.



acceptedStatus (L2) is the status of the winning opinion. The taxon profile shows that status **and the opinion/reference that produced it** — so "doubtful" always comes with "according to whom, and when" (FONC-720, CONS-300, CONS-310).

6. Occurrences, collections & geography

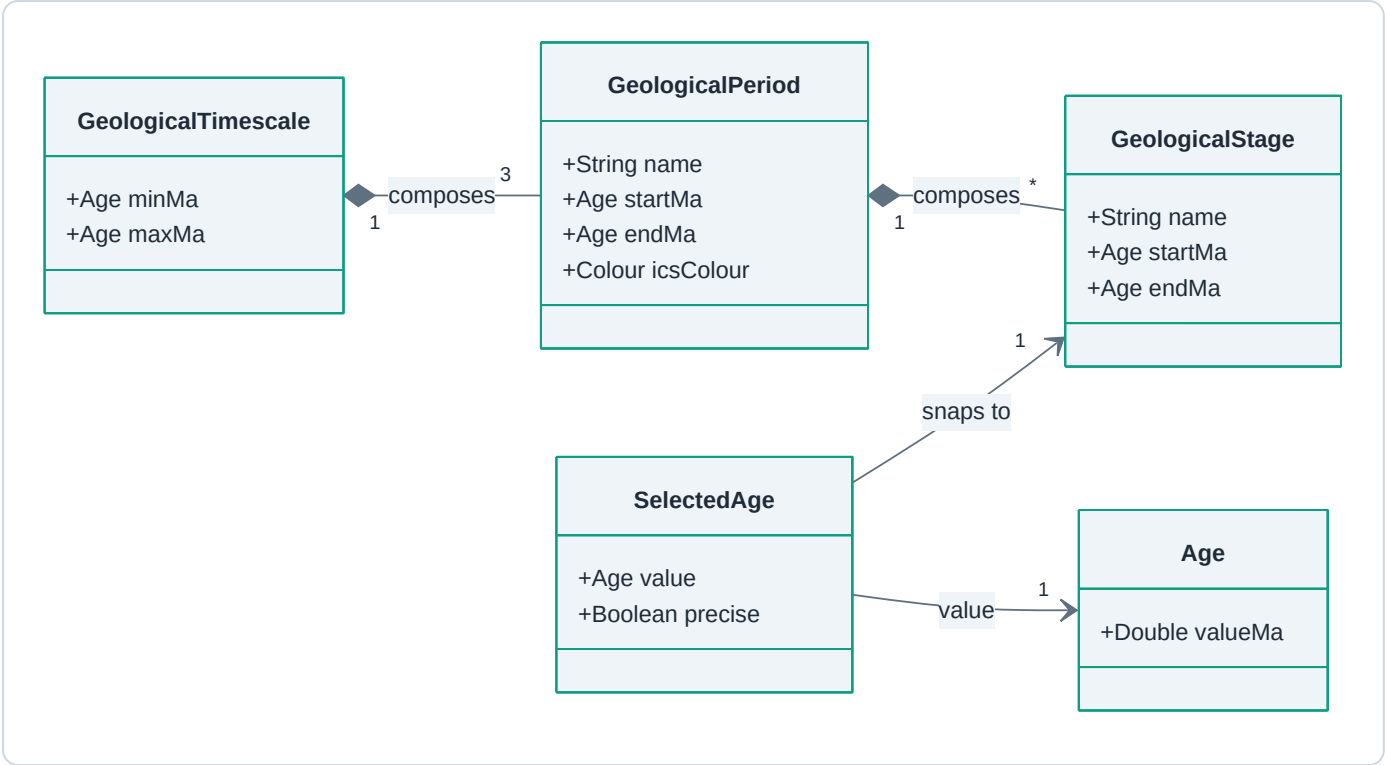
Geography and time belong to the **collection** (a locality), and the **occurrence** is a taxon *identified in a collection per a reference* — which is what PBDB actually models. The old EvidenceType enum is dropped: "discovery evidence, not a range" is a **display disclaimer** (FONC-1150), not a stored field.



- `PaleogeographicPosition.rotationModel` records *which* reconstruction produced it — so reconstructed is inherent to the field, and CONS-110/120/130 are satisfied structurally. (PBDB returns paleocoordinates; we store the model name.)
- `TimeRange.spansMultipleStages` feeds the derived approximate flag (FONC-1140, CONS-210).
- The occurrence panel presents its collection's position/time/source for the selected occurrence; the normalization above is what backs that view.

7. Geological time

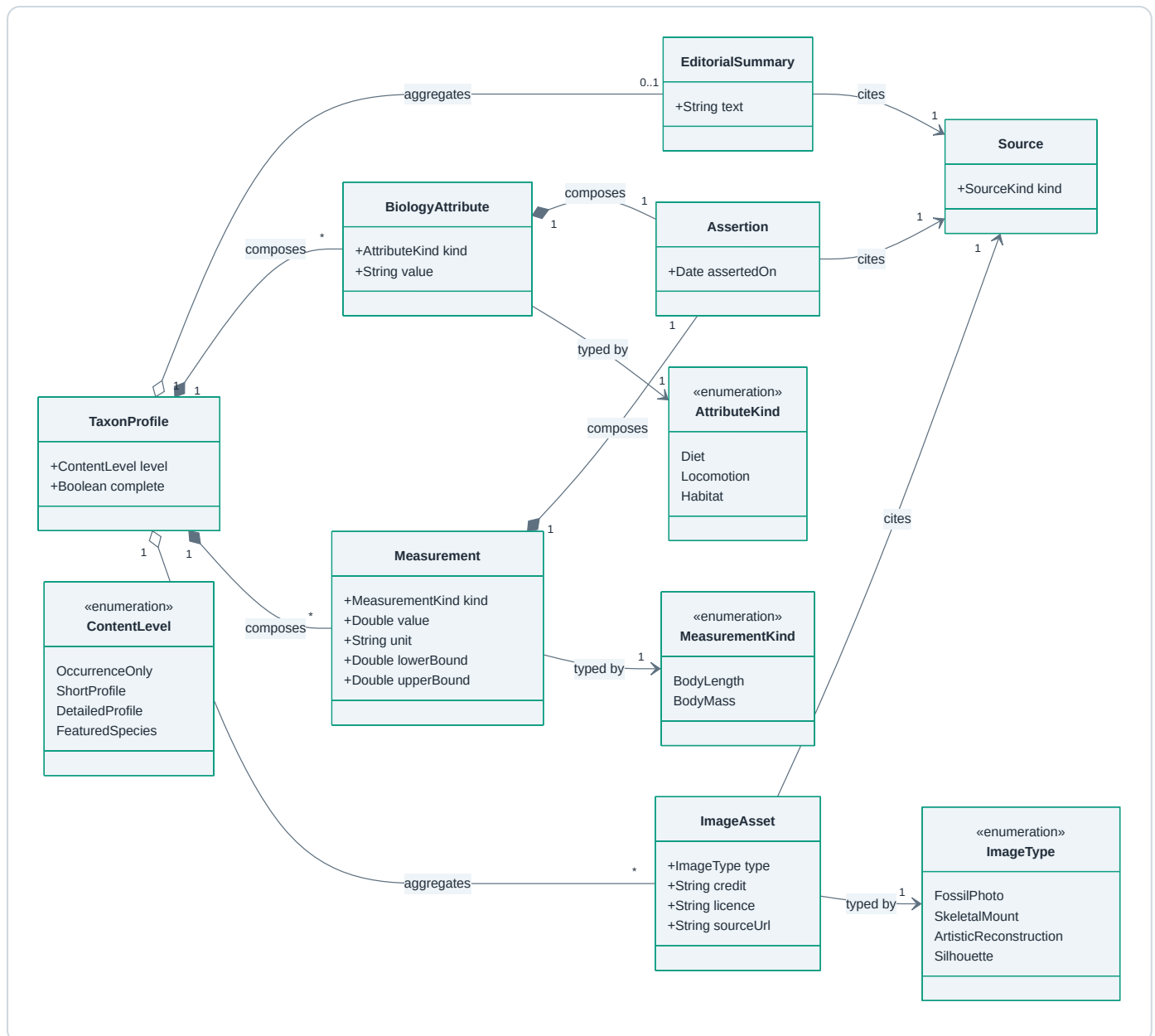
A fixed reference table (ICS), not sourced per-row — the timescale is standard.



SelectedAge snaps to GeologicalStage bakes the stage-stepped time control (§1.2) into the model.

8. Taxon profile & media (structured, sourced, sparse-by-design)

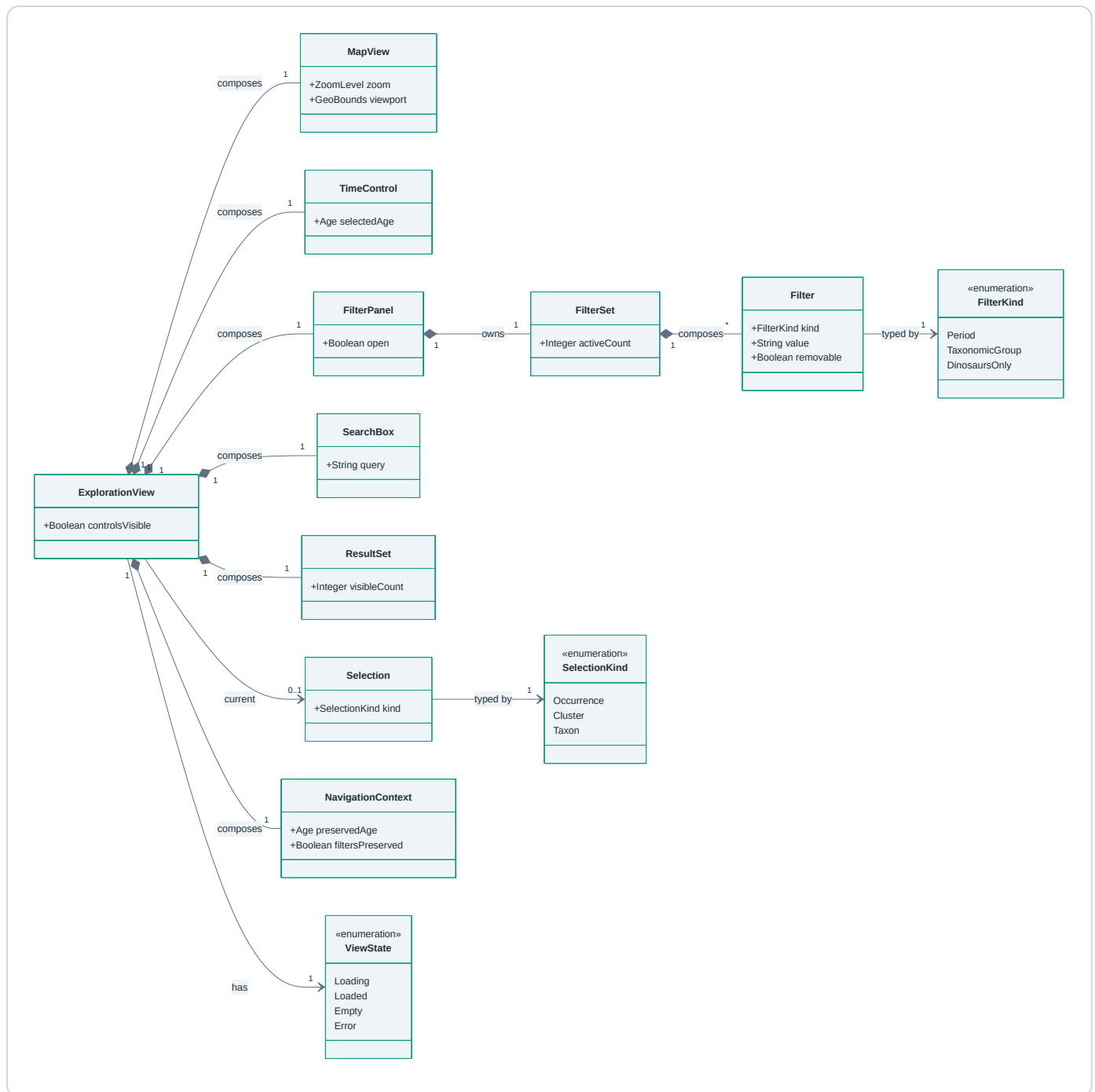
Biology is a set of **typed, nullable, sourced** attributes/measurements — not a free-text bag and not a generic key/value store. Sparsity is expected and handled by the explicit-missing rule and the content-level ladder.



- Each `BiologyAttribute` / `Measurement` composes an `Assertion` that cites a `Source` — so diet/mass carry "according to whom". Values may come from a PBDB measurement (primary), a Wikipedia figure (tertiary), or an editorial synthesis; the source kind decides whether the UI marks it *interpretative*.
- `Measurement` carries `lowerBound/upperBound` — uncertainty comes from the source, satisfying CONS-080/410 without inventing a confidence.
- `ContentLevel` is **derived** from how many of these are populated (+ presence of a summary/image), which is exactly the completeness ladder of FONC-430...470.
- `ImageAsset` carries `licence/credit/sourceUrl` and an `ImageType`, covering FONC-1190...1240.

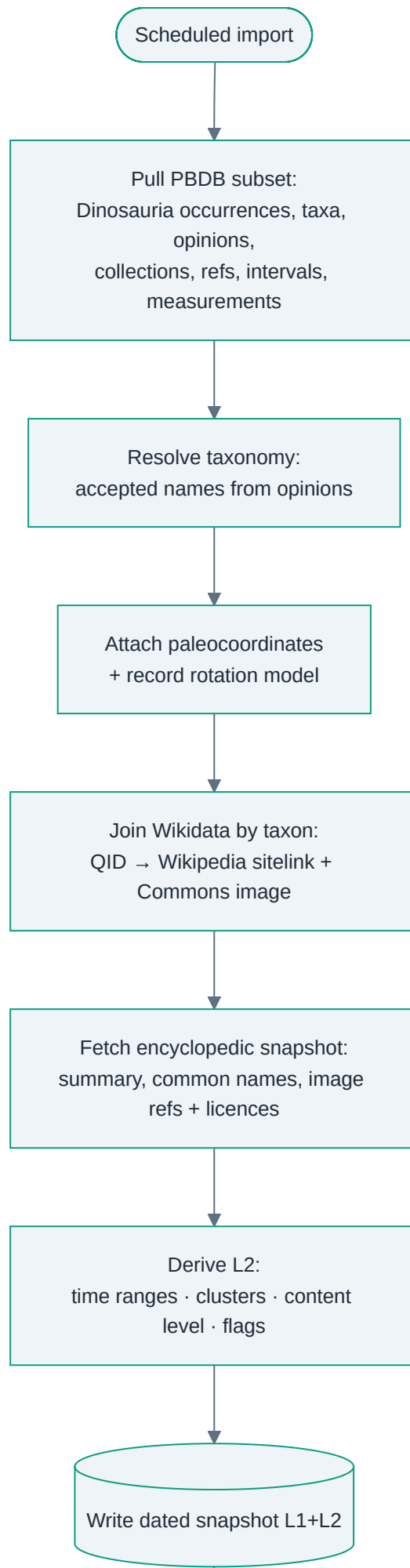
8b. Application & view model

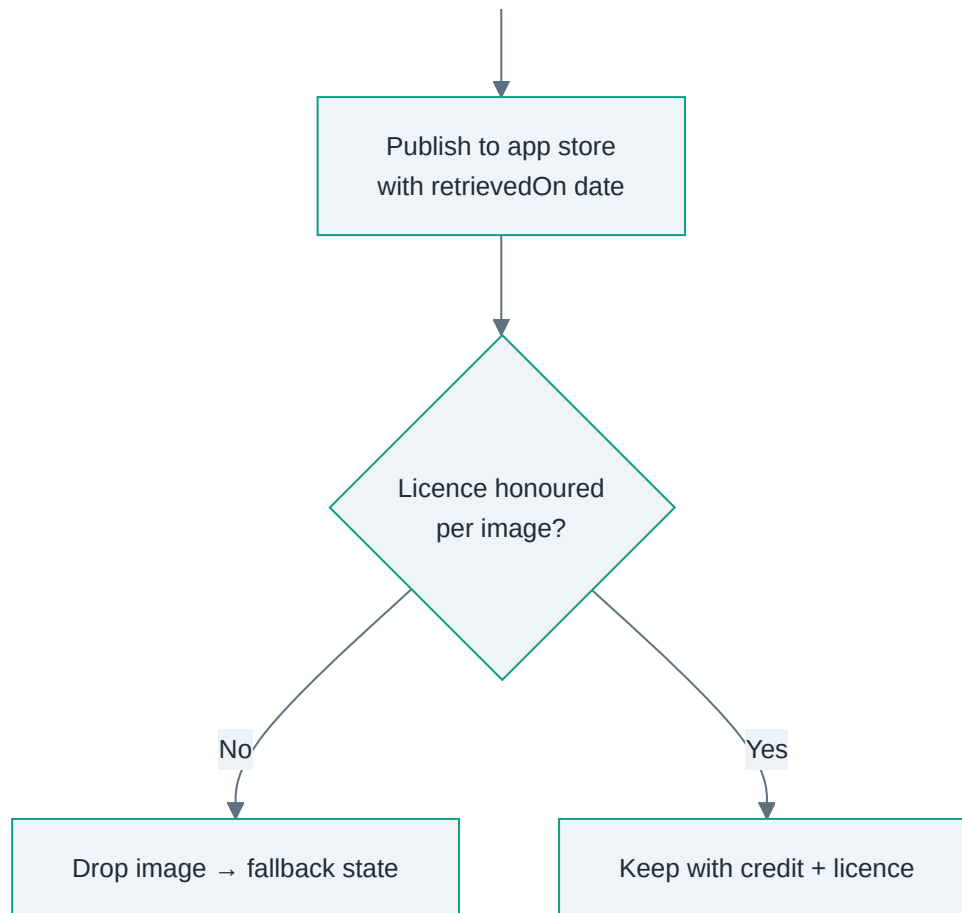
The runtime/view layer that the interface composes. These enums (`ViewState`, `FilterKind`, `SelectionKind`) are intrinsic — a view genuinely is one of loading/loaded/empty/error — unlike the data-layer classifications above.



NavigationContext is what preserves the selected age and active filters across navigation and load failures (FONC-1010/1020/1340).

9. Ingestion pipeline





Snapshot, not live. The app serves from our own dated store; it does not query PBDB/Wikipedia at request time. This is deliberate: it makes performance predictable (PERF-010...060), makes the atlas resilient to upstream downtime (PERF-280...310), and gives every field an honest, stable import date. The cost is a periodic re-import job, which is far cheaper than the alternatives.

10. Risks & mitigations

Risk	Mitigation
Becoming a paleobiology authority (taxonomy, dating, reconstruction)	Don't. Consume PBDB's accepted names, intervals, and paleocoords; store the rotation-model name for honesty.
PBDB ↔ Wikipedia mismatching	Join by Wikidata QID , not name; unmatched taxa degrade gracefully to "no encyclopedic content".
Wikipedia licensing / attribution	Store per-image <code>licence/credit/sourceUrl</code> ; attribute + share-alike summary text; drop anything we can't honour.
Tertiary content read as fact	<code>SourceKind</code> = <code>Encyclopedic</code> ; UI marks it <i>interpretative</i> ; never equal editorial status to PBDB (CONS-100).
Sparse profiles	Designed state: explicit "not available" (FONC-490) + content-level ladder; not a bug.
Upstream drift / downtime	Dated snapshots; app never calls upstream live.

11. How this satisfies the requirements

Requirement area	Mechanism here
Identifiable source per occurrence/time range (FONC-1100, PERF-140/150, CONS-390/400)	Every L1 row carries a Source; the app can't render without one
Distinguish primary / database / editorial (CONS-420)	SourceKind + Source.derivedFrom chain
Fossil-derived vs interpretative (FONC-670, FONC-1110, CONS-440)	Derived interpretative flag from Source.kind; assertions never mixed with facts in one field
Reconstructed / approximate marked (FONC-1130/1140, CONS-110/210)	Derived flags from rotationModel and spansMultipleStages
Missing shown explicitly (FONC-490, FONC-1120, PERF-180/190)	Nullable typed fields + derived missing flag
Validity/doubtful flagged with basis (FONC-720, CONS-300/310)	TaxonomicOpinion + winning opinion shown with its reference
Content levels (FONC-430...470)	Derived ContentLevel from populated structured fields
Import/consultation date, external links (FONC-1170/1180, CONS-430)	Source.retrievedOn; snapshot date; Source.url
Numeric values need source/uncertainty (CONS-080/410)	Measurement bounds + cited Source; no invented confidence
Images typed & credited (FONC-1200...1240)	ImageAsset.type/credit/licence/sourceUrl

12. Open design decisions (for a future increment)

- **Import cadence & diffing** — full re-snapshot vs. incremental; how to surface "updated since last import".
- **Paleocoordinate model** — accept PBDB's default rotation model, or pin a specific one (e.g. a named GPlates model) and record it per position.
- **Editorial featured-species workflow** — where L3 prose is authored and how it is reviewed (ties to the spec-first workflow).
- **Search index** — scientific-name search (MVP) and later common-name search (FONC-770) over the encyclopedic layer.

This document names PBDB + Wikipedia/Wikidata as the concrete sources; the **functional specification** stays source-neutral by design (it requires *an identifiable source*, not a particular provider).

Appendix A — Requirements index

Derived, searchable catalog of every requirement in the [functional specification](#). It introduces no new requirements; summaries are abbreviations of the normative text. If a summary conflicts with the specification, the specification wins. IDs and priorities are canonical.

Legend — Priority: MVP / V1 / V2. Every ID here exists in the specification. Verbs (must/should) are preserved in the source; the summaries drop them for brevity.

Features (FONC-*)

ID	Priority	Section	Short summary
FONC-010	MVP	Main exploration view	Provide a main view combining map, time control and filters.
FONC-020	MVP	Main exploration view	Dinosaurs are the active category by default on first load.
FONC-030	MVP	Main exploration view	Give access to an occurrence, a taxon page and a filter panel.
FONC-040	MVP	Main exploration view	Permanently display the selected geological age.
FONC-050	MVP	Main exploration view	Permanently display the selected taxonomic group.
FONC-060	MVP	Main exploration view	Permanently display the count of visible occurrences.
FONC-070	V1	Main exploration view	Show a short textual summary of the current selection.
FONC-080	MVP	Main exploration view	Allow active filters to be reset from the main view.
FONC-090	MVP	Time exploration	Allow exploration of the 252–66 Ma interval.
FONC-100	MVP	Time exploration	Divide the interval into Triassic, Jurassic, Cretaceous.
FONC-110	MVP	Time exploration	Display the selected age in Ma.
FONC-120	MVP	Time exploration	Change the selected age via a visible time control.
FONC-130	MVP	Time exploration	Update the map when the selected age changes.
FONC-140	MVP	Time exploration	Update visible occurrences when the selected age changes.
FONC-150	MVP	Time exploration	Show an occurrence only when the age overlaps its time range.
FONC-160	MVP	Time exploration	Show a taxon only when the age overlaps its time range.
FONC-170	MVP	Time exploration	Visually distinguish a precise age from a time range.
FONC-180	V1	Time exploration	Show the geological stage for the selected age when available.
FONC-190	V1	Time exploration	Allow quick selection of a complete period.
FONC-200	V2	Time exploration	Compare two geological ages side by side.
FONC-210	MVP	Paleogeographic map	Display a world map for the selected age.
FONC-220	MVP	Paleogeographic map	Show approximate continental positions for the age.
FONC-230	MVP	Paleogeographic map	Display occurrences as points or grouped points.
FONC-240	MVP	Paleogeographic map	Distinguish individual occurrences from groups.
FONC-250	MVP	Paleogeographic map	Allow the user to zoom the map.
FONC-260	MVP	Paleogeographic map	Allow the user to pan the map horizontally and vertically.
FONC-270	MVP	Paleogeographic map	Allow a displayed occurrence to be selected.

ID	Priority	Section	Short summary
FONC-280	MVP	Paleogeographic map	Show an information panel when an occurrence is selected.
FONC-290	MVP	Paleogeographic map	Panel shows taxon, time range, modern & paleo position, source.
FONC-300	MVP	Paleogeographic map	Clearly indicate the map is a paleogeographic reconstruction.
FONC-310	V1	Paleogeographic map	Allow a present-day continents comparison layer.
FONC-320	V1	Paleogeographic map	Allow occurrence display to be toggled on/off.
FONC-330	V2	Paleogeographic map	Animate the map transition between two ages.
FONC-340	MVP	Scope of groups	Treat non-avian dinosaurs as the main content.
FONC-350	MVP	Scope of groups	Allow only non-avian dinosaurs to be displayed.
FONC-360	MVP	Scope of groups	Distinguish dinosaurs from other included Mesozoic reptiles.
FONC-370	MVP	Scope of groups	Cover major dinosaur groups when data is available.
FONC-380	V1	Scope of groups	Include selected secondary reptile groups.
FONC-390	V1	Scope of groups	Allow only included secondary groups to be displayed.
FONC-400	MVP	Scope of groups	Do not present the app as a complete atlas of all Mesozoic life.
FONC-410	MVP	Scope of groups	Indicate whether a taxon is main or secondary content.
FONC-420	V1	Scope of groups	Allow secondary groups to be hidden.
FONC-430	MVP	Content levels	Classify each displayed taxon by content level.
FONC-440	MVP	Content levels	Support an "Occurrence only" level.
FONC-450	MVP	Content levels	Support a "Short profile" level.
FONC-460	V1	Content levels	Support a "Detailed profile" level.
FONC-470	V1	Content levels	Support a "Featured species" level.
FONC-480	MVP	Content levels	Clearly indicate when a taxon profile is incomplete.
FONC-490	MVP	Content levels	Show unavailable fields with an explicit label, not blank.
FONC-500	V1	Content levels	Filter or sort taxa by content level.
FONC-510	MVP	Taxon profile	Provide a profile for each selectable taxon.
FONC-520	MVP	Taxon profile	Display the scientific name on the profile.
FONC-530	MVP	Taxon profile	Display the taxonomic rank on the profile.
FONC-540	MVP	Taxon profile	Display the available taxonomic classification.
FONC-550	MVP	Taxon profile	Display the known time range of the taxon.
FONC-560	MVP	Taxon profile	Display the known fossil occurrences of the taxon.
FONC-570	MVP	Taxon profile	Display the modern discovery locations.
FONC-580	MVP	Taxon profile	Display the reconstructed paleogeographic positions.
FONC-590	MVP	Taxon profile	Display the sources used for main profile information.
FONC-600	V1	Taxon profile	Display assumed diet when available.
FONC-610	V1	Taxon profile	Display assumed locomotion mode when available.
FONC-620	V1	Taxon profile	Display estimated length when available.
FONC-630	V1	Taxon profile	Display estimated mass when available.
FONC-640	V1	Taxon profile	Display a short summary for detailed/featured taxa.
FONC-650	V1	Taxon profile	Display related taxa when available.

ID	Priority	Section	Short summary
FONC-660	V1	Taxon profile	Display taxa from the same formation/region and overlapping time.
FONC-670	MVP	Taxon profile	Indicate when information is interpretative, not fossil-derived.
FONC-680	MVP	Classification	Allow taxa to be explored by taxonomic group.
FONC-690	MVP	Classification	Display the taxonomic hierarchy of a selected taxon.
FONC-700	MVP	Classification	Select a group and update the map accordingly.
FONC-710	MVP	Classification	Distinguish taxonomic levels (clade, family, genus, species).
FONC-720	MVP	Classification	Flag invalid/doubtful/synonymous/uncertain taxa when known.
FONC-730	V1	Classification	Navigate from a species to its parent groups.
FONC-740	V1	Classification	Navigate from a group to its genera or species.
FONC-750	V1	Classification	Filter the map from a selection in the taxonomic tree.
FONC-760	MVP	Search & filters	Search a taxon by scientific name.
FONC-770	V1	Search & filters	Search a taxon by common/popular name when available.
FONC-780	MVP	Search & filters	Filter occurrences by geological period.
FONC-790	MVP	Search & filters	Filter occurrences by taxonomic group.
FONC-800	MVP	Search & filters	Filter to display only non-avian dinosaurs.
FONC-810	V1	Search & filters	Filter occurrences by diet.
FONC-820	V1	Search & filters	Filter occurrences by modern discovery region.
FONC-830	V1	Search & filters	Filter occurrences by taxon content level.
FONC-840	V1	Search & filters	Filter by data completeness or reliability level.
FONC-850	MVP	Search & filters	Display the number of results matching active filters.
FONC-860	MVP	Search & filters	Show an explicit empty state when no result matches.
FONC-870	MVP	Search & filters	Remove an active filter individually.
FONC-880	MVP	Search & filters	Remove all active filters in a single action.
FONC-890	MVP	Occurrences & formations	Provide a panel/profile for each selected occurrence.
FONC-900	MVP	Occurrences & formations	Display the modern location of an occurrence when available.
FONC-910	MVP	Occurrences & formations	Display the reconstructed paleo position when available.
FONC-920	MVP	Occurrences & formations	Display the time range of an occurrence.
FONC-930	MVP	Occurrences & formations	Display the source of an occurrence.
FONC-940	V1	Occurrences & formations	Display the formation name of an occurrence when available.
FONC-950	V1	Occurrences & formations	Provide a profile for a selected geological formation.
FONC-960	V1	Occurrences & formations	Display taxa known from a selected formation.
FONC-970	V1	Occurrences & formations	Display the age/time range of a formation when available.
FONC-980	V1	Occurrences & formations	Compare modern vs paleo position of a location.
FONC-990	MVP	Navigation	Open a taxon profile from a selected occurrence.
FONC-1000	MVP	Navigation	Return to the map from a taxon profile.
FONC-1010	MVP	Navigation	Preserve the selected age when navigating to a profile.
FONC-1020	MVP	Navigation	Preserve active filters when navigating to a profile.
FONC-1030	V1	Navigation	Open a formation profile from a selected occurrence.

ID	Priority	Section	Short summary
FONC-1040	V1	Navigation	Open a taxon profile from a formation profile.
FONC-1050	V1	Navigation	Open a related taxon profile from a taxon profile.
FONC-1060	V1	Navigation	Return to the previous search result after a profile.
FONC-1070	MVP	Navigation	Reach a taxon profile in ≤ 2 actions from a visible occurrence.
FONC-1080	MVP	Navigation	Return to the map in ≤ 1 action from a taxon profile.
FONC-1090	MVP	Sources & uncertainty	Display the sources of the scientific data used.
FONC-1100	MVP	Sources & uncertainty	Display an identifiable source per visible occurrence.
FONC-1110	MVP	Sources & uncertainty	Distinguish fossil-derived data from interpretative data.
FONC-1120	MVP	Sources & uncertainty	Indicate missing info instead of an unmarked assumption.
FONC-1130	MVP	Sources & uncertainty	Indicate when a geographic position is reconstructed.
FONC-1140	MVP	Sources & uncertainty	Indicate when a time range is approximate.
FONC-1150	MVP	Sources & uncertainty	Show occurrences as discovery evidence, not distribution.
FONC-1160	V1	Sources & uncertainty	Show a confidence level for interpretative info when available.
FONC-1170	V1	Sources & uncertainty	Show the consultation/import date of data when available.
FONC-1180	V1	Sources & uncertainty	Show a link to the external source when available.
FONC-1190	V1	Images	Show an image for detailed/featured taxa when available.
FONC-1200	V1	Images	Distinguish fossil photo, artistic reconstruction, silhouette.
FONC-1210	V1	Images	Display the source/credit for an image when shown.
FONC-1220	V1	Images	Indicate when an image is an artistic representation.
FONC-1230	V1	Images	Do not present art as evidence of exact appearance.
FONC-1240	MVP	Images	Show an alternative state when no image is available.
FONC-1250	V2	Images	Compare animal size with a human reference when available.
FONC-1260	MVP	Interface states	Show a loading state while the map initially loads.
FONC-1270	MVP	Interface states	Show a loading state when opening a not-yet-available profile.
FONC-1280	MVP	Interface states	Show an empty state when filters return no occurrence.
FONC-1290	MVP	Interface states	Show an empty state when a search returns no taxon.
FONC-1300	MVP	Interface states	Show a message when a profile exists but has minimal data.
FONC-1310	MVP	Interface states	Show a clear error when a map cannot be loaded.
FONC-1320	MVP	Interface states	Show a clear error when a taxon profile cannot be loaded.
FONC-1330	MVP	Interface states	Allow a failed load to be retried.
FONC-1340	MVP	Interface states	Preserve active filters after a loading failure.
FONC-1350	V2	Comparison & guided paths	Compare two geological periods in a dedicated view.
FONC-1360	V2	Comparison & guided paths	Compare two taxonomic groups in a dedicated view.
FONC-1370	V1	Comparison & guided paths	Display a timeline for a selected taxon.
FONC-1380	V1	Comparison & guided paths	Show a taxon's first and last appearance on a timeline.
FONC-1390	V2	Comparison & guided paths	Select a region and list its taxa for the selected age.
FONC-1400	V2	Comparison & guided paths	Provide thematic guided paths.
FONC-1410	V2	Comparison & guided paths	Provide a guided path on large theropods.

ID	Priority	Section	Short summary
FONC-1420	V2	Comparison & guided paths	Provide a guided path on Jurassic dinosaurs.
FONC-1430	V2	Comparison & guided paths	Provide a guided path on Late Cretaceous dinosaurs.

Constraints (CONS-*)

ID	Priority	Section	Short summary
CONS-010	MVP	Scientific scope	Limit main temporal scope to the Mesozoic (252–66 Ma).
CONS-020	MVP	Scientific scope	Exclude post-K–Pg species except as marked context.
CONS-030	MVP	Scientific scope	Treat non-avian dinosaurs as the main scope.
CONS-040	MVP	Scientific scope	Treat other Mesozoic reptiles as secondary scope.
CONS-050	MVP	Scientific scope	Do not cover all Mesozoic non-reptile life in main scope.
CONS-060	MVP	Scientific scope	Display only taxa with an identifiable source.
CONS-070	MVP	Scientific scope	Display only occurrences with an identifiable source.
CONS-080	MVP	Scientific scope	No size/mass/age value without source or uncertainty.
CONS-090	V1	Scientific scope	Limit secondary groups to an explicitly defined list.
CONS-100	MVP	Scientific scope	Avoid wording equating secondary content with dinosaurs.
CONS-110	MVP	Geographic representation	Distinguish modern from reconstructed coordinates.
CONS-120	MVP	Geographic representation	State ancient maps are reconstructions, not observations.
CONS-130	MVP	Geographic representation	Do not present fossil points as exact life positions.
CONS-140	MVP	Geographic representation	Present fossil points as discovery/collection locations.
CONS-150	MVP	Geographic representation	Do not extrapolate continuous distribution from points.
CONS-160	MVP	Geographic representation	No complete distribution area unless explicitly sourced.
CONS-170	V1	Geographic representation	Note insufficient geographic resolution for the map scale.
CONS-180	MVP	Geographic representation	Use consistent geographic units throughout.
CONS-190	MVP	Temporal representation	Represent ages in millions of years before present.
CONS-200	MVP	Temporal representation	Show time ranges with min and max boundaries when available.
CONS-210	MVP	Temporal representation	Clearly indicate a broad time range.
CONS-220	MVP	Temporal representation	Do not present a range as a precise age.
CONS-230	MVP	Temporal representation	Do not show non-overlapping taxa as contemporaneous.
CONS-240	V1	Temporal representation	Indicate period/epoch/stage for a time range when available.
CONS-250	MVP	Scientific consistency	Do not display a taxon outside its known time range.
CONS-260	MVP	Scientific consistency	Do not display an occurrence outside its known time range.
CONS-270	MVP	Scientific consistency	No predator-prey relationship as certain without a source.
CONS-280	MVP	Scientific consistency	Do not present assumed behavior as established fact.
CONS-290	MVP	Scientific consistency	Distinguish facts, estimates and hypotheses.
CONS-300	MVP	Scientific consistency	Flag controversial/invalid/doubtful/synonymous taxa when known.
CONS-310	V1	Scientific consistency	Note when a well-known taxon is not valid in the data used.
CONS-320	MVP	Editorial content	Use an informative, scientific, accessible tone.

ID	Priority	Section	Short summary
CONS-330	MVP	Editorial content	Avoid unverifiable sensationalist wording.
CONS-340	MVP	Editorial content	Avoid absolute claims under incomplete/uncertain data.
CONS-350	MVP	Editorial content	Use scientific names as the primary reference.
CONS-360	V1	Editorial content	Use common names as reading aids when available.
CONS-370	MVP	Editorial content	Display units for numeric values.
CONS-380	V1	Editorial content	Avoid an overly childish art direction.
CONS-390	MVP	Sources & provenance	Associate each visible occurrence with an identifiable source.
CONS-400	MVP	Sources & provenance	Associate each time range with a source or dataset.
CONS-410	MVP	Sources & provenance	Associate each size/mass estimate with source or uncertainty.
CONS-420	MVP	Sources & provenance	Distinguish primary source, database and editorial synthesis.
CONS-430	V1	Sources & provenance	Show the consultation/import date of an external source.
CONS-440	MVP	Sources & provenance	Do not mix sourced data and unmarked assumptions in a field.
CONS-450	MVP	Interface & usability	Keep time, map and filter controls visible on the main view.
CONS-460	MVP	Interface & usability	Reach a taxon profile in ≤ 2 actions from a visible occurrence.
CONS-470	MVP	Interface & usability	Return to the map in ≤ 1 action from a taxon profile.
CONS-480	MVP	Interface & usability	Keep nomenclature consistent for periods, groups, taxa.
CONS-490	MVP	Interface & usability	Do not hide interpretation-changing uncertainty behind extra steps.
CONS-500	V1	Interface & usability	Support desktop and tablet screens.
CONS-510	V2	Interface & usability	Support mobile phones without loss of main features.

Performance (PERF - *)

ID	Priority	Section	Short summary
PERF-010	MVP	Response time	Display the main view within 5 s after initial load.
PERF-020	MVP	Response time	Display first useful content within 3 s after initial load.
PERF-030	MVP	Response time	Update visible occurrences within 1 s after an age change (data loaded).
PERF-040	MVP	Response time	Open a taxon profile within 1 s when data is already loaded.
PERF-050	MVP	Response time	Show a loading indicator when an update exceeds 500 ms.
PERF-060	MVP	Response time	Give visual feedback within 100 ms after zoom/pan.
PERF-070	V1	Response time	Perform a simple text search within 500 ms (data loaded).
PERF-080	MVP	Readability & density	Keep main labels readable at ≥ 12 CSS px.
PERF-090	MVP	Readability & density	Prevent > 30 markers overlapping in 100×100 px without grouping.
PERF-100	MVP	Readability & density	Cluster occurrences when density blocks reliable selection.
PERF-110	MVP	Readability & density	Provide ≥ 3 zoom levels: global, regional, local.
PERF-120	MVP	Readability & density	Make each point/cluster selectable at $\geq 24 \times 24$ CSS px.
PERF-130	V1	Readability & density	Show names only when zoom avoids excessive overlap.
PERF-140	MVP	Visible data quality	Show a source for 100% of visible occurrences.
PERF-150	MVP	Visible data quality	Show a time range for 100% of visible occurrences.

ID	Priority	Section	Short summary
PERF-160	MVP	Visible data quality	Show minimum classification for 100% of visible taxa.
PERF-170	MVP	Visible data quality	Show ≥1 parent group above genus/species for 100% of taxa when available.
PERF-180	MVP	Visible data quality	Explicitly signal missing fields in 100% of affected profiles.
PERF-190	MVP	Visible data quality	No profile with >20% silently empty fields.
PERF-200	V1	Visible data quality	Provide ≥50 detailed dinosaur profiles in first enriched version.
PERF-210	V1	Visible data quality	Provide ≥10 featured species in first enriched version.
PERF-220	MVP	Accessibility	Allow the main view to be used with a mouse or trackpad.
PERF-230	MVP	Accessibility	Allow the main filters to be used with a keyboard.
PERF-240	MVP	Accessibility	Maintain sufficient contrast for main text.
PERF-250	MVP	Accessibility	Convey critical information without relying only on color.
PERF-260	V1	Accessibility	Provide alternative/descriptive text for species images.
PERF-270	MVP	Accessibility	Allow selection without precision below 24×24 CSS px.
PERF-280	MVP	Display robustness	Show a clear error when map data cannot be loaded.
PERF-290	MVP	Display robustness	Show a clear error when a profile cannot be loaded.
PERF-300	MVP	Display robustness	Allow retry of a failed load without full app reload.
PERF-310	MVP	Display robustness	Preserve active filters after a load failure.
PERF-320	MVP	Display robustness	Show an empty state when filters return no occurrence.
PERF-330	MVP	Display robustness	Show an empty state when search returns no result.
PERF-340	MVP	MVP validation	Complete the period → filter → occurrence → profile → map scenario.
PERF-350	MVP	MVP validation	Complete the search → profile → occurrences-on-map scenario.
PERF-360	MVP	MVP validation	Complete the age-change → occurrence-update scenario without full reload.
PERF-370	MVP	MVP validation	Complete the empty-filter → empty-state → reset scenario.

Appendix B — Requirements traceability

Derived document. It maps requirements from the [functional specification](#) to the documentation artifacts that will describe or verify them. It introduces no new requirements.

Purpose. Give every requirement a forward path to a screen, a UML diagram, and an acceptance criterion. Diagram names are **placeholders** until the source files exist (see [../uml/README.md](#)). Acceptance criteria IDs point to [acceptance-criteria.md](#); TBD marks criteria not yet written (V1/V2).

Status legend: Draft — mapped, not yet verified. Update to Specified / Designed / Verified as artifacts are produced.

Screen names refer to pages under [../mockups/](#): Exploration view, Taxon profile, Occurrence panel, Filters panel, Empty/error states. Constraints and performance requirements are often cross-cutting and are grouped by subsection; their "Related screen" is "All / cross-cutting" where they apply everywhere.

Features (FONC-*)

Requirement ID	Related screen	Related UML diagram	Acceptance criteria	Status
FONC-010	Exploration view	Use case diagram, exploration-flow activity	AC-010	Draft
FONC-020	Exploration view, Filters panel	Exploration-flow activity	AC-020	Draft
FONC-030	Exploration view	Use case diagram, exploration-flow activity	AC-010	Draft
FONC-040	Exploration view	Exploration-flow activity	AC-010	Draft
FONC-050	Exploration view	Exploration-flow activity	AC-010	Draft
FONC-060	Exploration view, Filters panel	Filter-application activity	AC-010, AC-120	Draft
FONC-070	Exploration view	Exploration-flow activity	TBD	Draft
FONC-080	Exploration view, Filters panel	Filter-application activity	AC-130	Draft
FONC-090	Exploration view	Age-change activity	AC-040	Draft
FONC-100	Exploration view	Age-change activity	AC-040	Draft
FONC-110	Exploration view	Age-change activity	AC-040	Draft
FONC-120	Exploration view	Age-change activity, map-loading state	AC-030	Draft
FONC-130	Exploration view	Age-change activity, map-loading state	AC-030	Draft
FONC-140	Exploration view	Age-change activity	AC-030	Draft
FONC-150	Exploration view	Age-change activity, domain model	AC-030	Draft
FONC-160	Exploration view, Taxon profile	Age-change activity, domain model	AC-030	Draft
FONC-170	Exploration view	Age-change activity	AC-040	Draft
FONC-180	Exploration view	Age-change activity	TBD	Draft
FONC-190	Exploration view	Age-change activity	TBD	Draft
FONC-200	(V2 comparison view)	—	TBD	Draft
FONC-210	Exploration view	Map-loading state, domain model	AC-050	Draft
FONC-220	Exploration view	Domain model	AC-050	Draft

Requirement ID	Related screen	Related UML diagram	Acceptance criteria	Status
FONC-230	Exploration view	Domain model	AC-060	Draft
FONC-240	Exploration view	Domain model	AC-060	Draft
FONC-250	Exploration view	—	AC-070	Draft
FONC-260	Exploration view	—	AC-070	Draft
FONC-270	Exploration view, Occurrence panel	Occurrence-selection activity	AC-080	Draft
FONC-280	Occurrence panel	Occurrence-selection activity	AC-080	Draft
FONC-290	Occurrence panel	Occurrence-selection activity, domain model	AC-080	Draft
FONC-300	Exploration view	—	AC-050	Draft
FONC-310	Exploration view	—	TBD	Draft
FONC-320	Exploration view, Filters panel	—	TBD	Draft
FONC-330	Exploration view	—	TBD	Draft
FONC-340	Exploration view, Filters panel	Domain model	AC-020, AC-230	Draft
FONC-350	Filters panel	Filter-application activity	AC-020	Draft
FONC-360	Exploration view, Taxon profile	Domain model	AC-230	Draft
FONC-370	Filters panel	Domain model	AC-230	Draft
FONC-380	Filters panel	Domain model	TBD	Draft
FONC-390	Filters panel	—	TBD	Draft
FONC-400	Exploration view	—	AC-230	Draft
FONC-410	Taxon profile, Occurrence panel	Domain model	AC-230	Draft
FONC-420	Filters panel	—	TBD	Draft
FONC-430	Taxon profile	Domain model	AC-170	Draft
FONC-440	Taxon profile	Domain model	AC-170	Draft
FONC-450	Taxon profile	Domain model	AC-170	Draft
FONC-460	Taxon profile	Domain model	TBD	Draft
FONC-470	Taxon profile	Domain model	TBD	Draft
FONC-480	Taxon profile	Taxon-profile-loading state	AC-170, AC-240	Draft
FONC-490	Taxon profile	—	AC-170	Draft
FONC-500	Filters panel	—	TBD	Draft
FONC-510	Taxon profile	Taxon-profile-flow activity	AC-100	Draft
FONC-520	Taxon profile	—	AC-100	Draft
FONC-530	Taxon profile	Domain model	AC-100	Draft
FONC-540	Taxon profile	Domain model	AC-100	Draft
FONC-550	Taxon profile	Domain model	AC-100	Draft
FONC-560	Taxon profile	Domain model	AC-100	Draft
FONC-570	Taxon profile	Domain model	AC-100	Draft
FONC-580	Taxon profile	Domain model	AC-100	Draft
FONC-590	Taxon profile	Domain model	AC-100, AC-180	Draft
FONC-600	Taxon profile	—	TBD	Draft

Requirement ID	Related screen	Related UML diagram	Acceptance criteria	Status
FONC-610	Taxon profile	—	TBD	Draft
FONC-620	Taxon profile	—	TBD	Draft
FONC-630	Taxon profile	—	TBD	Draft
FONC-640	Taxon profile	—	TBD	Draft
FONC-650	Taxon profile	—	TBD	Draft
FONC-660	Taxon profile	Domain model	TBD	Draft
FONC-670	Taxon profile	—	AC-180	Draft
FONC-680	Exploration view, Filters panel	Domain model	AC-220	Draft
FONC-690	Taxon profile	Domain model	AC-220	Draft
FONC-700	Exploration view, Filters panel	Filter-application activity	AC-220	Draft
FONC-710	Taxon profile	Domain model	AC-220	Draft
FONC-720	Taxon profile	Domain model	AC-220	Draft
FONC-730	Taxon profile	—	TBD	Draft
FONC-740	Taxon profile	—	TBD	Draft
FONC-750	Filters panel	—	TBD	Draft
FONC-760	Exploration view	Search state	AC-150	Draft
FONC-770	Exploration view	Search state	TBD	Draft
FONC-780	Filters panel	Filter-application activity	AC-120	Draft
FONC-790	Filters panel	Filter-application activity	AC-120	Draft
FONC-800	Filters panel	Filter-application activity	AC-020, AC-120	Draft
FONC-810	Filters panel	—	TBD	Draft
FONC-820	Filters panel	—	TBD	Draft
FONC-830	Filters panel	—	TBD	Draft
FONC-840	Filters panel	—	TBD	Draft
FONC-850	Exploration view, Filters panel	Filter-application activity	AC-120	Draft
FONC-860	Empty/error states, Filters panel	Empty-state activity, filter state	AC-140	Draft
FONC-870	Filters panel	Filter-application activity, filter state	AC-130	Draft
FONC-880	Filters panel	Filter-application activity, filter state	AC-130	Draft
FONC-890	Occurrence panel	Occurrence-selection activity	AC-080	Draft
FONC-900	Occurrence panel	Domain model	AC-080	Draft
FONC-910	Occurrence panel	Domain model	AC-080	Draft
FONC-920	Occurrence panel	Domain model	AC-080	Draft
FONC-930	Occurrence panel	Domain model	AC-080, AC-180	Draft
FONC-940	Occurrence panel	Domain model	TBD	Draft
FONC-950	(Formation profile — future)	Domain model	TBD	Draft
FONC-960	(Formation profile — future)	Domain model	TBD	Draft
FONC-970	(Formation profile — future)	Domain model	TBD	Draft
FONC-980	Occurrence panel	Domain model	TBD	Draft

Requirement ID	Related screen	Related UML diagram	Acceptance criteria	Status
FONC-990	Occurrence panel, Taxon profile	Taxon-profile-flow activity	AC-090	Draft
FONC-1000	Taxon profile	Taxon-profile-flow activity	AC-110	Draft
FONC-1010	Taxon profile	Taxon-profile-flow activity	AC-110	Draft
FONC-1020	Taxon profile	Taxon-profile-flow activity	AC-110	Draft
FONC-1030	Occurrence panel	—	TBD	Draft
FONC-1040	(Formation profile — future)	—	TBD	Draft
FONC-1050	Taxon profile	—	TBD	Draft
FONC-1060	Taxon profile	Search state	TBD	Draft
FONC-1070	Occurrence panel, Taxon profile	Taxon-profile-flow activity	AC-090	Draft
FONC-1080	Taxon profile	Taxon-profile-flow activity	AC-110	Draft
FONC-1090	Taxon profile, Occurrence panel	—	AC-180	Draft
FONC-1100	Occurrence panel	—	AC-180	Draft
FONC-1110	Taxon profile	—	AC-180	Draft
FONC-1120	Taxon profile	—	AC-170, AC-180	Draft
FONC-1130	Occurrence panel, Taxon profile	—	AC-180	Draft
FONC-1140	Occurrence panel, Taxon profile	—	AC-180	Draft
FONC-1150	Occurrence panel	—	AC-180	Draft
FONC-1160	Taxon profile	—	TBD	Draft
FONC-1170	Taxon profile, Occurrence panel	—	TBD	Draft
FONC-1180	Taxon profile, Occurrence panel	—	TBD	Draft
FONC-1190	Taxon profile	—	TBD	Draft
FONC-1200	Taxon profile	—	TBD	Draft
FONC-1210	Taxon profile	—	TBD	Draft
FONC-1220	Taxon profile	—	TBD	Draft
FONC-1230	Taxon profile	—	TBD	Draft
FONC-1240	Taxon profile, Empty/error states	Taxon-profile-loading state	AC-190	Draft
FONC-1250	Taxon profile	—	TBD	Draft
FONC-1260	Exploration view, Empty/error states	Map-loading state	AC-200	Draft
FONC-1270	Taxon profile, Empty/error states	Taxon-profile-loading state	AC-200	Draft
FONC-1280	Empty/error states	Empty-state activity, filter state	AC-140	Draft
FONC-1290	Empty/error states	Search state	AC-160	Draft
FONC-1300	Taxon profile, Empty/error states	Taxon-profile-loading state	AC-240	Draft
FONC-1310	Empty/error states	Map-loading state	AC-210	Draft
FONC-1320	Empty/error states	Taxon-profile-loading state	AC-210	Draft
FONC-1330	Empty/error states	Map-loading state, taxon-profile-loading state	AC-210	Draft
FONC-1340	Empty/error states, Filters panel	Filter state	AC-210	Draft
FONC-1350	(V2 comparison view)	—	TBD	Draft
FONC-1360	(V2 comparison view)	—	TBD	Draft

Requirement ID	Related screen	Related UML diagram	Acceptance criteria	Status
FONC-1370	Taxon profile	—	TBD	Draft
FONC-1380	Taxon profile	—	TBD	Draft
FONC-1390	(V2 region view)	—	TBD	Draft
FONC-1400	(V2 guided paths)	—	TBD	Draft
FONC-1410	(V2 guided paths)	—	TBD	Draft
FONC-1420	(V2 guided paths)	—	TBD	Draft
FONC-1430	(V2 guided paths)	—	TBD	Draft

Constraints (CONS-*) — grouped by subsection

Constraints are cross-cutting. They are grouped here; each row lists the ID range and the primary artifacts that must honor it.

Requirement IDs	Related screen	Related UML diagram	Acceptance criteria	Status
CONS-010...CONS-100 (Scientific scope)	Exploration view, Taxon profile	Domain model	AC-180, AC-230	Draft
CONS-110...CONS-180 (Geographic representation)	Exploration view, Occurrence panel	Domain model	AC-050, AC-180	Draft
CONS-190...CONS-240 (Temporal representation)	Exploration view, Taxon profile, Occurrence panel	Age-change activity, domain model	AC-040, AC-180	Draft
CONS-250...CONS-310 (Scientific consistency)	Taxon profile, Occurrence panel	Age-change activity, domain model	AC-180	Draft
CONS-320...CONS-380 (Editorial content)	All / cross-cutting	—	AC-180	Draft
CONS-390...CONS-440 (Sources & provenance)	Taxon profile, Occurrence panel	Domain model	AC-180	Draft
CONS-450...CONS-510 (Interface & usability)	Exploration view, Taxon profile	Exploration-flow activity, taxon-profile-flow activity	AC-010, AC-090, AC-110	Draft

Performance (PERF-*) — grouped by subsection

Requirement IDs	Related screen	Related UML diagram	Acceptance criteria	Status
PERF-010...PERF-070 (Response time)	Exploration view, Taxon profile	Map-loading state, taxon-profile-loading state	AC-010, AC-030, AC-200	Draft
PERF-080...PERF-130 (Readability & density)	Exploration view	—	AC-060, AC-070	Draft
PERF-140...PERF-210 (Visible data quality)	Exploration view, Taxon profile, Occurrence panel	Domain model	AC-170, AC-180	Draft
PERF-220...PERF-270 (Accessibility)	Exploration view, Filters panel	—	AC-070, AC-120	Draft
PERF-280...PERF-330 (Display robustness)	Empty/error states, Filters panel	Map-loading state, taxon-profile-loading state, filter state	AC-140, AC-160, AC-210	Draft

Requirement IDs	Related screen	Related UML diagram	Acceptance criteria	Status
PERF-340 (Validation scenario)	Exploration view → Occurrence panel → Taxon profile	Exploration-flow activity, taxon-profile-flow activity	AC-090, AC-110	Draft
PERF-350 (Validation scenario)	Exploration view → Taxon profile	Taxon-profile-flow activity, search state	AC-150	Draft
PERF-360 (Validation scenario)	Exploration view	Age-change activity	AC-030	Draft
PERF-370 (Validation scenario)	Filters panel, Empty/error states	Filter-application activity, empty-state activity	AC-140, AC-130	Draft

Appendix C — Acceptance criteria

Derived document. Testable acceptance criteria (AC) derived from MVP requirements in the [functional specification](#). ACs restate requirements as verifiable Given/When/Then scenarios; they do not introduce new requirements. If an AC conflicts with the specification, the specification wins.

Numbering: ACs are numbered in increments of 10. Each AC lists the requirement IDs it verifies. Coverage focuses on **MVP** requirements; a small number of V1/V2 placeholders (AC - 9xx) are stubbed for future use.

Resolved dependencies: criteria that once depended on [open questions](#) now reflect their resolutions — e.g. an "action" is defined per OQ-040, the time control steps by geological stage per OQ-030, and MVP content is dinosaurs-only per OQ-050. Notes below point to the governing decision.

AC-010 — Main exploration view loads

Related requirements: FONC-010, FONC-030, FONC-040, FONC-050, FONC-060, CONS-450, PERF-010, PERF-020.

- **Given** the application is opened on a standard desktop connection,
- **When** the main exploration view finishes loading,
- **Then** the map, the time control, and the filter controls are all visible and remain visible; the selected geological age, the selected taxonomic group, and the count of visible occurrences are each shown; first useful content appears within 3 s and the full view within 5 s.

AC-020 — Dinosaurs are the default active category

Related requirements: FONC-020, FONC-340, FONC-350, FONC-800.

- **Given** the main exploration view loads for the first time,
- **When** no filter has been changed by the user,
- **Then** non-avian dinosaurs are the active category, and only non-avian dinosaur occurrences are shown.

AC-030 — Changing the selected age updates the map and occurrences

Related requirements: FONC-120, FONC-130, FONC-140, FONC-150, FONC-160, PERF-030, PERF-360.

- **Given** the exploration view is loaded and the required data is already loaded,
- **When** the user changes the selected age via the time control,
- **Then** the map redraws for the new age and the visible occurrences update to only those whose time range overlaps the new age, within 1 s and without a full application reload.

AC-040 — Geological age is displayed correctly

Related requirements: FONC-090, FONC-100, FONC-110, FONC-170, CONS-190.

- **Given** the exploration view is loaded,
- **When** the user inspects the time control,

- **Then** the selected age is shown in Ma within the 252–66 Ma interval; the interval is divided into Triassic, Jurassic, and Cretaceous; and a selected precise age is visually distinct from a displayed time range.

Note: per OQ-030 (resolved), the time control steps by geological stage and displays the selected age in Ma.

AC-050 — Map is clearly a paleogeographic reconstruction

Related requirements: FONC-210, FONC-220, FONC-300, CONS-120.

- **Given** a selected age,
- **When** the world map is displayed,
- **Then** it shows the approximate continental positions for that age and carries a clear indication that it is a scientific paleogeographic reconstruction, not a direct observation.

AC-060 — Occurrences render as points and clusters

Related requirements: FONC-230, FONC-240, PERF-090, PERF-100, PERF-110.

- **Given** occurrences exist for the selected age and filters,
- **When** they are displayed on the map,
- **Then** individual occurrences are visually distinct from occurrence groups; when density would place more than 30 markers in a 100×100 CSS px area they are clustered; and at least 3 zoom levels (global, regional, local) are available.

AC-070 — Map zoom and pan with accessible targets

Related requirements: FONC-250, FONC-260, PERF-060, PERF-120, PERF-270.

- **Given** the map is displayed,
- **When** the user zooms or pans,
- **Then** the map responds with visual feedback within 100 ms, and every point or cluster remains selectable within an interactive area of at least 24×24 CSS px.

AC-080 — Selecting an occurrence opens its information panel

Related requirements: FONC-270, FONC-280, FONC-290, FONC-890, FONC-900, FONC-910, FONC-920, FONC-930.

- **Given** an occurrence is visible on the map,
- **When** the user selects it,
- **Then** an information panel opens showing at minimum the associated taxon, the time range, the modern location, the paleogeographic position, and the source.

AC-090 — Opening a taxon profile from an occurrence

Related requirements: FONC-990, FONC-1070, CONS-460, PERF-340.

- **Given** an occurrence is visible on the map,

- **When** the user opens the associated taxon profile,
- **Then** the taxon profile opens in no more than 2 actions from the visible occurrence.

Note: per OQ-040 (resolved), an "action" is one primary interaction that causes a navigation/state transition (click/tap or keyboard equivalent); hover/scroll/zoom/pan are not counted. So occurrence → info panel (1) → open profile (2) satisfies the ≤2-action limit.

AC-100 — Taxon profile shows required content

Related requirements: FONC-510, FONC-520, FONC-530, FONC-540, FONC-550, FONC-560, FONC-570, FONC-580, FONC-590.

- **Given** a taxon selected from the map or search results,
- **When** its profile opens,
- **Then** the profile shows the scientific name, taxonomic rank, available classification, known time range, known fossil occurrences, modern discovery locations, reconstructed paleogeographic positions, and the sources used.

AC-110 — Returning to the map preserves state

Related requirements: FONC-1000, FONC-1010, FONC-1020, FONC-1080, CONS-470.

- **Given** the user navigated from the map to a taxon profile with a selected age and active filters,
- **When** the user returns to the map,
- **Then** the return takes no more than 1 action, and the previously selected age and active filters are still applied.

Note: "action" is defined per OQ-040 (resolved) — see AC-090.

AC-120 — Filtering by period, group, and dinosaurs-only with a live count

Related requirements: FONC-780, FONC-790, FONC-800, FONC-850, FONC-060.

- **Given** the exploration view is loaded,
- **When** the user filters by geological period, by taxonomic group, or to dinosaurs only,
- **Then** the visible occurrences update to match, and the displayed result count reflects the active filters.

AC-130 — Removing filters individually and all at once

Related requirements: FONC-870, FONC-880, FONC-080.

- **Given** one or more filters are active,
- **When** the user removes a single filter, or resets all filters,
- **Then** the removed filter(s) no longer apply, the result set and count update accordingly, and a single action clears all active filters.

AC-140 — Empty state when filters return no occurrence

Related requirements: FONC-860, FONC-1280, PERF-320, PERF-370.

- **Given** active filters that match no occurrence,
- **When** the filtered result is displayed,
- **Then** an explicit empty state is shown (not a blank map), and the user can reset the filters to leave the empty state.

AC-150 — Searching a taxon by scientific name

Related requirements: FONC-760, PERF-350.

- **Given** the exploration view is loaded,
- **When** the user searches for a taxon by its scientific name,
- **Then** matching taxa are returned, and from a result the user can open the profile and display the taxon's occurrences on the map.

AC-160 — Empty state when a search returns nothing

Related requirements: FONC-1290, PERF-330.

- **Given** a search term that matches no taxon,
- **When** the search completes,
- **Then** an explicit empty state is shown for the search.

AC-170 — Content level and incomplete-profile labeling

Related requirements: FONC-430, FONC-440, FONC-450, FONC-480, FONC-490, FONC-1120, PERF-180, PERF-190.

- **Given** a displayed taxon,
- **When** its profile or listing is shown,
- **Then** the taxon carries a content level (at least "Occurrence only" or "Short profile"); an incomplete profile is clearly flagged; every unavailable field is shown with an explicit label rather than a blank; and no profile has more than 20% silently empty fields.

AC-180 — Sources and uncertainty are always visible

Related requirements: FONC-670, FONC-1090, FONC-1100, FONC-1110, FONC-1130, FONC-1140, FONC-1150, CONS-390, CONS-440.

- **Given** any visible occurrence or taxon profile,
- **When** its data is displayed,
- **Then** an identifiable source is shown; interpretative data is distinguished from fossil-derived data; reconstructed positions and approximate time ranges are labeled as such; occurrences are presented as discovery evidence rather than distribution boundaries; and sourced data is not mixed with unmarked assumptions in the same field.

AC-190 — Alternative state when no image is available

Related requirements: FONC-1240.

- **Given** a taxon with no available image,
- **When** its profile is displayed,
- **Then** an explicit alternative (fallback) state is shown in place of the image.

AC-200 — Loading states for map and profile

Related requirements: FONC-1260, FONC-1270, PERF-050.

- **Given** the map is loading initially, or a taxon profile whose data is not already available is being opened,
- **When** the operation takes longer than 500 ms,
- **Then** a loading state / indicator is displayed until content is ready.

AC-210 — Error, retry, and filter preservation

Related requirements: FONC-1310, FONC-1320, FONC-1330, FONC-1340, PERF-280, PERF-290, PERF-300, PERF-310.

- **Given** a map or taxon profile fails to load,
- **When** the failure is surfaced,
- **Then** a clear error message is shown; the user can retry without reloading the entire application; and any active filters are preserved through the failure and retry.

AC-220 — Taxonomy exploration and group selection

Related requirements: FONC-680, FONC-690, FONC-700, FONC-710, FONC-720.

- **Given** the exploration view,
- **When** the user explores by taxonomic group and selects one,
- **Then** the map updates for that group; a selected taxon shows its available taxonomic hierarchy with distinguished levels (clade, family, genus, species); and invalid/doubtful/synonymous/uncertain taxa are flagged when that information is available.

AC-230 — Scope labeling: main vs secondary content

Related requirements: FONC-360, FONC-400, FONC-410.

- **Given** displayed taxa,
- **When** their content is shown,
- **Then** non-avian dinosaurs are distinguished from other Mesozoic reptiles, each taxon is marked as main or secondary content, and the application does not present itself as a complete atlas of all Mesozoic life.

Note: per OQ-050 (resolved), the MVP data set is dinosaurs-only, so this criterion verifies the labeling **mechanism** (every taxon tagged "main content") rather than the presence of secondary taxa. The mechanism must be ready for secondary groups in V1 (FONC-380/FONC-390).

AC-240 — Minimal-data profile message

Related requirements: FONC-480, FONC-1300.

- **Given** a taxon profile that exists but contains only minimal data,
 - **When** it is opened,
 - **Then** an explicit message indicates that the profile is minimal/incomplete.
-

Placeholders for later releases (V1 / V2)

These are stubs to be expanded when the corresponding requirements are scheduled. Do not treat them as complete.

- **AC-900 — Selection summary text (V1):** covers FONC-070. *TODO*.
- **AC-910 — Present-day comparison layer (V1):** covers FONC-310, FONC-980. *TODO*.
- **AC-920 — Enriched profile fields (V1):** covers FONC-600...FONC-640. *TODO*.
- **AC-930 — Formation profiles (V1):** covers FONC-940...FONC-970, FONC-1030, FONC-1040. *TODO*.
- **AC-940 — Image typing and credit (V1):** covers FONC-1190...FONC-1230, PERF-260. *TODO*.
- **AC-950 — Taxon timeline (V1):** covers FONC-1370, FONC-1380. *TODO*.
- **AC-960 — Period/group comparison views (V2):** covers FONC-200, FONC-1350, FONC-1360. *TODO*.
- **AC-970 — Guided paths (V2):** covers FONC-1400...FONC-1430. *TODO*.
- **AC-980 — Mobile support (V2):** covers CONS-510. *TODO*.

Appendix D — Glossary

Derived document. Terms and definitions are taken from the [functional specification](#) §0.3. If a definition here conflicts with the specification, the specification wins.

The specification glossary is reproduced below. A second block lists additional domain terms used across the product documentation and UML domain model; these are **documentation aids**, not new requirements.

Specification glossary (§0.3)

Term	Definition
Mesozoic	Geological interval between approximately 252 Ma and 66 Ma.
Ma	Millions of years before present.
Non-avian dinosaurs	Dinosaurs in the common sense, excluding modern birds and strictly avian lineages.
Taxon	Biological classification unit, for example clade, family, genus or species.
Fossil occurrence	Documented fossil evidence associated with a taxon, a discovery location and a geological time range.
Geological formation	Named geological unit associated with rocks and, in some cases, fossils.
Modern position	Present-day location where a fossil was discovered.
Paleogeographic position	Approximate reconstructed position of that location during a past geological period.
Time range	Interval between a minimum age boundary and a maximum age boundary.
Interpretative data	Inferred or synthesized information, for example diet, estimated mass or assumed behavior.
Sourced data	Information linked to an identifiable source.

Supporting terms (documentation aids)

These terms appear in the specification body or in the domain-model documentation and are defined here for convenience. They do not extend the product scope.

Term	Definition
Action	One primary user interaction that causes a navigation or state transition — a click/tap or its keyboard equivalent. Hover, scroll, zoom, and pan do not count. Used to measure the navigation limits FONC-1070/CONS-460 (≤2 actions) and FONC-1080/CONS-470 (≤1 action). (Resolves OQ-040.)
Content level	Classification of how much information a taxon has: Occurrence only, Short profile, Detailed profile, or Featured species (see FONC-430...FONC-470).
Occurrence only	Content level for taxa that only have basic fossil occurrence data (FONC-440).
Short profile	Content level for taxa with a name, classification, time range and at least one occurrence (FONC-450).
Detailed profile	Content level for taxa with sourced morphological or ecological information (FONC-460).
Featured species	Content level for species with an enriched profile, a dedicated map and associated information (FONC-470).
Taxonomic group	A grouping used to explore and filter taxa, e.g. theropods, sauropodomorphs, ornithopods, thyreophorans, ceratopsians, pachycephalosaurus (FONC-370).
Taxonomic rank	The level of a taxon in the classification hierarchy, e.g. clade, family, genus, species (FONC-530, FONC-710).
Main content	Non-avian dinosaurs; the primary editorial scope (FONC-340, CONS-030).

Term	Definition
Secondary content	Selected other Mesozoic reptiles (e.g. pterosaurs, marine reptiles, crocodylomorphs, Mesozoic lepidosaurs), clearly labeled as secondary (FONC-380, CONS-040).
Geological period	One of the three Mesozoic periods: Triassic, Jurassic, Cretaceous (FONC-100).
Geological stage	A finer subdivision of geological time associated with an age, when available (FONC-180, CONS-240).
Source	An identifiable origin for data; may be a primary source, a database, or an editorial synthesis (CONS-390, CONS-420).
Fossil fact / estimate / hypothesis	The three evidence categories the system must distinguish (CONS-290).
CSS px	A CSS reference pixel, the unit used for readability and interactive-area thresholds (PERF-080, PERF-120).